MAST20026 - Real Analysis – Semester 1, 2025

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Subject Information:	On the Canvas LMS – Check!	
Lectures:	Wednesday 11am-12pm	Lyle Theatre
	Thursday 3-4pm	Lyle Theatre
	Friday 3-4pm	Lyle Theatre
Assessments:	5 written assignments (20% total) Weekly quizzes (10%) Final exam (70%)	

How to use these lecture slides

These slides are designed to facilitate the lecture presentation and note taking.

The full content of the course is not presented on these slides. Rather, plenty of blank space is provided for the lecturer to write out content and draw pictures during the lecture.

Students are strongly encouraged to attend/watch lectures regularly and take notes in the blank space provided.

Additionally, students should study and full digest the course material after each lecture. It may help to write out the main definitions and theorems from lecture and come up with relevant examples and counter-examples.

This subject has two main aims:

- To study how to write **mathematical proofs**;
- To study **Real Analysis**.

One important topic in Real Analysis is Fourier series.

A Fourier series takes the form

$$f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(kx) + b_k \sin(kx))$$

where x is a variable and a_k and b_k are real number constants.

Fourier series have many important applications in mathematics and physics.
For instance, solutions to the heat equation in thermal dynamics can be represented as:

$$f(x,t) = \sum_{k=1}^{\infty} b_k \, \mathrm{e}^{\mathrm{kt}} \sin(kx)$$

Another example is the Weierstrass function

$$f(x) = \sum_{k=1}^{\infty} \frac{1}{2^k} \sin(3^k \pi x)$$

It is continuous at every real number x but differentiable nowhere!

Chapter 1:
Proof and Mathematical Logic

1.1 Propositional Logic

Propositional logic is about variables and functions which take *two values*, called "True" and "False".

Definition 1.1. *A* <u>*statement*</u> *is a sentence or expression that is either true or false.*

A statement takes on the role of a logical variable (taking values **true** or **false**). We generally use lower case letters $p, q, r, \ldots$ to represent statements.

Some books use the word <u>proposition</u> statements rather than propositions.

Example. (Which of these are statements?)

- $1 + 1 = 3$
- $f(x)$ is continuous.
- 6 is the largest integer.
- Potato
- If it is Saturday, there is no MAST20026 lecture.

Example. (Which of these are statements?)

- $x > 2$

- For every $x \in \mathbb{Z}$, $x^2 \geq 0$.

- Every even number greater than 2 is the sum of two primes.

- $\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ is invertible and $1 + 1 = 3$.

Connectives. We can combine statements to give new statements, called compound statements. They are constructed from:

1. primitive statements (i.e. logical variables)
2. connectives
3. parentheses (to remove ambiguity)

Often, we use upper case letters $A, B, C, \ldots$ to denote compound statements.

The connectives we will consider are:

- negation
- conjunction
- disjunction
- implication
- biconditional

Definition 1.2. *Let p be a statement. The* <u>*negation*</u> *of p is the statement "not p" and is denoted by $\sim p$. The truth value of $\sim p$ is the opposite of p.*

Examples

Let p be the statement "$5 > 0$".

Let r be the statement "this lecture is boring".

Definition 1.3. *Let p and q be statements. The* <u>*conjunction*</u> *of p and q is the statement "p and q" and is denoted by $p \wedge q$. The truth value of $p \wedge q$ is given by the following truth table.*

Example.

Let p be the statement "$\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ is invertible." Let q be the statement $1 + 1 = 3$.

In the English language, there are two different usages of the word "*or*".

The *exclusive* use of "or":

I will ride my bike or catch the train.

usually means that one or the other of these modes of transportation is used, but not both. This is **not** the mathematical usage of "or".

Definition 1.4. *Let p and q be statements. The* <u>*disjunction*</u> *of p and q is the statement "p or q" and is denoted by $p \vee q$. The truth value of $p \vee q$ is given by the following truth table.*

Disjunction is an *inclusive* use of "or"; that is, it allows for the possibility that both statements are true.

Example

Let p be the statement "$\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ is invertible." Let q be the statement $1 + 1 = 3$.

Example 1.5. For which values of p and q are the following compound statements true:

(a) $(\sim p \wedge q) \wedge (p \vee q)$

(b) $(p \vee \sim p) \wedge (q \vee \sim q)$

Definition 1.9. *Let p and q be statements. The* <u>*implication*</u> *"p implies q" and is denoted by $p \Rightarrow q$. The truth value of $p \Rightarrow q$ is given by the following truth table.*

Example. Let p be the statement

"The temperature outside is colder than 0°C."

And let q be the statement

"Brian wears a beanie when he is outside."

Definition 1.10. *Let p and q be statements. The* <u>*converse*</u> *of the implication $p \Rightarrow q$ is the implication $q \Rightarrow p$.*

In general, knowing $p \Rightarrow q$ tells us nothing about the truth value of the converse.

Definition 1.11. *Let p and q be statements. The* <u>*biconditional*</u> *"p if and only q" is denoted by $p \Leftrightarrow q$. The truth value of $p \Leftrightarrow q$ is given by the following truth table.*

Example.

$$A \text{ is invertible if and only if } \det A \neq 0$$

Definition 1.12. *Let p and q be statements. The* <u>*contrapositive*</u> *of the implication $p \Rightarrow q$ is the implication $(\sim q) \Rightarrow (\sim p)$.*

Example.

If it is Saturday, then there is no MAST20026 lecture.

Definition 1.7. *Let A be a compound statement. We say A is a* <u>*tautology*</u> *when A is always true regardless of the truth values of the simpler statements used to build A.*

Definition 1.8. *Let A be a compound statement. We say A is a* <u>*contradiction*</u> *when A is always false regardless of the truth values of the simpler statements used to build A.*

Definition 1.13. *Let r and s be statements. We say r and s are* <u>*logically equivalent*</u> *when $r \Leftrightarrow s$ is a tautology. When r and s are logically equivalent we write $r \equiv s$.*

Exercise 1.14. Prove $\sim (p \vee q) \equiv (\sim p) \wedge (\sim q)$.

1.2 First-Order Logic

Consider the following:

- $f(x)$ is continuous and differentiable

- $x^2 \geq 9$

- The set A has ten elements.

Mathematical sentences whose truth depend on variables (mathematical objects) are called <u>conditions</u>. The three examples above are conditions.

In all likelihood, you have seen conditions before, though perhaps not explicitly.

Conditions are hiding in the background when we use set-builder notation.

For example, consider the set

$$\{x|x^2 \geq 9\}$$

When we consider a condition for a particular value in a domain, the result is a statement. Thus we can combine conditions using the connectives. For instance:

$$\sim p(x) \qquad p(x) \vee q(x) \qquad p(x) \wedge q(x)$$

$$p(x) \Rightarrow q(x) \qquad p(x) \Leftrightarrow q(x)$$

Example.

$p(A)$: "A is invertible" $\qquad$ $q(A)$: "$\det(A) \neq 0$"

Truth values partition the domain into those for which the corresponding statement is *true* and those for which the corresponding statement is *false*.

Let q be a condition. There are four (overlapping) possibilities for this partition:

1. $q(x)$ is true for every x in the domain.
2. $q(x)$ is true for at least one x in the domain.
3. $q(x)$ is false for every x in the domain.
4. $q(x)$ is false for at least one x in the domain.

Example. Let $\mathcal{P}_1$ denote the set of all polynomials of degree one

$$\mathcal{P}_1 = \{f \mid f(x) = ax + b, \quad a \neq 0\}$$

For $f \in \mathcal{P}_1$, let $p(f)$ be the condition

$$f(x) \text{ crosses the } x\text{-axis}$$

On the other hand, for $f \in \mathcal{P}_1$, the condition

$$q : f'(x) > 0$$

Definition 1.15. *Let* $p(x)$ be a condition over a domain D.

The statement "$p(x)$ is true for every x in the domain D" is denoted as

$$(\forall x \in D)\ p(x).$$

We refer to the symbol $\forall$ as the <u>universal quantifier</u>, and we say $(\forall x \in D)\ p(x)$ is a <u>universally quantified statement</u>.

The statement "$p(x)$ is true for at least one x in the domain D" is denoted as

$$(\exists x \in D)\ p(x).$$

We refer to the symbol $\exists$ as the <u>existential quantifier</u>, and we say $(\exists x \in D)\ p(x)$ is an <u>existential quantified statement</u>.

Negation of quantifiers

$$\sim [(\exists x \in D)\ \ p(x)] \quad \equiv \quad (\forall x \in D)\ \ \sim p(x)$$

Why?

Similarly

$$\sim [(\forall x \in D)\ \ p(x)] \quad \equiv \quad (\exists x \in D)\ \ \sim p(x)$$

Example 1.16. Express the following statement in the language of formal logic.

There exists a natural number n so that $2n$ is odd.

Example.

Every vector space of dimension 3 has a basis.

Example.

There exists a vector space of dimension 3 with no basis.

Example 1.18.

For every $x \neq 0$ and every $y \neq 0$, the product of x and y is not zero.

Example. Let $p(f, g)$ be the condition "$g(x)$ is the first derivative of $f(x)$"

For instance, $p(x^2, 2x)$ is

For instance, $p(x^3 + 1, 2x)$ is

Fix $f(x) = x^2 + 1$. Then

We know from Calculus that for every polynomial $f(x)$, its derivative is another polynomial. Hence

Example. Consider the condition $q : x + 1 > y$ and the statement

$$(\forall x \in \mathbb{R})[(\exists y \in \mathbb{R})\ q(x, y)]$$

Now consider the statement

$$(\exists x \in \mathbb{R})[(\forall y \in \mathbb{R})\ q(x, y)]$$

Aside. Which of the following is easier for you to understand?

- $\because (\exists A^{-1} \Leftrightarrow \det A \neq 0) \wedge \det A = 0 \quad \therefore \sim (\exists A^{-1})$
- Recall that A^{-1} exists if and only if $\det A \neq 0$. Since $\det A = 0$, it then follows that A^{-1} does not exist.

Some definitions from number theory

Definition. Let n be an integer.

- We say n is <u>odd</u> if $(\exists k \in \mathbb{Z})\ \ n = 2k + 1$.
- We say n is <u>even</u> if $(\exists k \in \mathbb{Z})\ \ n = 2k$.

Definition. Let n be an integer and let d be a positive integer.
We say d <u>divides</u> n, and write $d|n$, if $(\exists k \in \mathbb{Z})\ \ n = dk$.

Definition. Let $p > 1$ be an integer. We say p is <u>prime</u> if the only positive integers d which divides p are $p = 1$ and $p = d$.

Definition. Let n be an integer. Let d and r be integers with $d > 0$ and $0 \leq r < d$.
We say <u>n divided by d gives a remainder r</u> if

$$(\exists k \in \mathbb{Z})\ \ n = dk + r.$$

Definition. Let x be a real number. We say x is <u>rational</u> if there exist integers p and $q \neq 0$ such that $x = p/q$. If x is not rational, then we say that x is <u>irrational</u>.

1.3 Mathematical Proofs

Most mathematical statements (i.e. theorems) that we encounter appear to be of the form

$$\text{If } p, \text{ then } q.$$

For example:

Theorem. *Let $f: \mathbb{R} \to \mathbb{R}$. If f is differentiable, then f is continuous.*

Theorem. *Let $f: \mathbb{R} \to \mathbb{R}$. If f is a quadratic function, then f crosses the x-axis at most twice.*

Theorem. *Let V be a vector space. If V is finite-dimensional, then V has a basis.*

These statements are in fact conditional statements depending on a variable:

$$(\forall f \in \mathcal{F})\ \ p(f) \Rightarrow q(f)$$

We want to show the following:

Theorem. $p \Rightarrow q$

Let's look at the truth table for $p \Rightarrow q$:

Definition 1.20. *A <u>mathematical proof</u> is a sequence of statements where each statement is:*

- *a declaration of notation,*
- *a premise, or*
- *a true statement that follows from:*
  - *a definition,*
  - *algebra, or*
  - *previous true statements.*

Definition 1.21. *A <u>formal mathematical proof</u> is a finite sequence of statements*

$$A_1, A_2, A_3, \dots, A_n$$

such that each A_i is either:

- *known or assumed true or*
- *an be inferred from a known or assumed true statement A_j with $j < i$ (i.e. a previous statement).*

A **good practical definition** of a mathematical proof is:

An argument in favor of a mathematical statement that will convince a preponderance of knowledgeable peers of its truth.

Why do we care about proofs?

- To communicate mathematics to our peers (and to ourselves).
- To verify the truth of mathematical claims and avoid errors.

Theorem 1.19. *Let x be an integer. If x is even, then x^2 is even.*

The hypothesis $p(x)$ is:

The conclusion $q(x)$ is:

Scrap work:

Theorem 1.19. *Let x be an integer. If x is even, then x^2 is even.*

Proof. (formal)

1.

2.

3.

4.

5.

6.

7.

Theorem 1.19. *Let x be an integer. If x is even, then x^2 is even.*

Proof. (semi-formal)

1.

2.

3.

4.

5.

6.

7.

Theorem 1.19. *Let x be an integer. If x is even, then x^2 is even.*

Proof. (informal)

Let x be an even integer. Since x is even there exists an integer k such that

$$x = 2k.$$

Computing, we find

$$x^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Thus there exists an integer ℓ such that $x^2 = 2\ell$; in particular, $\ell = 2k^2$. Therefore, x^2 is even. Therefore, if x is even, then x^2 is even.

Theorem 1.22. *Let x and y be integers. If x and y are even, then $(x+y)^2$ is even.*

The hypothesis is:

The conclusion is:

Scrap work:

Theorem 1.22. *Let x and y be integers. If x and y are even, then $(x+y)^2$ is even.*

Proof. (formal)

1.

2.

3.

4.

5.

6.

7.

8.

9.

10.

11.

12.

13.

Theorem 1.24. *Let* $x \in \mathbb{Z}$. *If* $x^2 - 6x + 5$ *is even, then* x *is odd.*

Attempted proof. (formal)

1.

2.

3.

4.

5.

6.

7.

Proof by Contraposition. Recall that

$$p \Rightarrow q \quad \equiv \quad \sim q \Rightarrow \sim p$$

Theorem 1.24. *Let* $x \in \mathbb{Z}$. *If* $x^2 - 6x + 5$ *is even, then* x *is odd.*

The contrapositive is:

Scrap work:

Theorem 1.24. *Let* $x \in \mathbb{Z}$. *If* $x^2 - 6x + 5$ *is even, then* x *is odd.*

Proof. We proceed by contraposition.

1.

2.

3.

4.

5.

6.

7.

Consider the following sequence of statements:

1. $\frac{1}{2}$ is an integer.	(premise)
2. $1 + \frac{1}{2}$ is an integer.	(1, integers are closed under addition)
3. $\frac{3}{2}$ is an integer.	(2)
4. There is an integer between 1 and 2.	(3)

Proof by contradiction.

Theorem 1.25. *Let* a *and* b *be real numbers. If* a *is a rational number and* b *is an irrational number, then the sum* $a + b$ *is irrational.*

To summarize what we have discussed, let's write out what generic direct proofs, proof by contraposition, and proofs by contradiction look like.

Theorem. $(\forall x \in D)\ \ p(x) \Rightarrow q(x)$

Direct proof.

1. Let $x \in D$ such that $p(x)$ is true. (premise)

2.

3.

4.

5. $q(x)$

6. $(\forall x \in D)\ \ p(x) \Rightarrow q(x)$ (1., 5.)

Theorem. $(\forall x \in D)\ \ p(x) \Rightarrow q(x)$

Proof by contraposition.

1. Let $x \in D$ such that $\sim q(x)$ is true. (premise)

2.

3.

4.

5. $\sim p(x)$

6. $(\forall x \in D)\ \ \sim q(x) \Rightarrow \sim p(x)$ (1., 5.)

7. $(\forall x \in D)\ \ p(x) \Rightarrow q(x)$ (contrapositive of 6.)

Theorem. p

Proof by contradiction.

1.	$\sim p$	(premise)
2.		
3.	r	
4.		
5.		
6.	$\sim r$	
7.	$r \wedge (\sim r)$	(3., 6.)

We can write proofs involving multiple cases using mini-direct proofs.

Theorem. *If a divided by* 8 *gives a remainder of either* 1 *or* 7*, then a^2 divided by* 8 *gives a remainder of* 1.

Proof. Let a be an integer such that a divided by 8 gives a remainder of 1 or 7. We shall consider the cases where the remainder is 1 or is 7 separately.

Case 1: a divided by 8 gives a remainder of 1.

1.	$(\exists n \in \mathbb{Z})\ a = 8n + 1$	(premise)
2.	???	(algebra)
3.	$(\exists m \in \mathbb{Z})\ a^2 = 8m + 1$	(3)

Case 2: a divided by 8 gives a remainder of 7.

1.	$(\exists n \in \mathbb{Z})\ a = 8n + 7$	(premise)
2.	???	(justification)
3.	$(\exists m \in \mathbb{Z})\ a^2 = 8m + 1$	(3)

Conclusion: Therefore, if a divided by 8 gives a remainder of either 1 or 7, then a^2 divided by 8 gives a remainder of 1.

Examples. The statement

$$(\exists x \in D)\, p(x)$$

is true precisely when $p(x)$ is true for at least one member of D.

Hence we can prove the statement by finding one a for which $p(a)$ is true.
We call such a value a an <u>example</u>.

Example.

There exists a linear function which passes through the origin.

Counter-examples. The statement

$$(\forall x \in D)\, p(x)$$

is true precisely when $p(x)$ is true for every member of D.

Hence we can disprove the statement by finding one a for which $p(a)$ is false. We call such a value a a <u>counter-example</u>.

Example.

Every prime number is odd.

Consider the following sum:

$$s(n) = \sum_{i=0}^{n} 2^i = 2^0 + 2^1 + 2^2 + \cdots + 2^n$$

When n is small, it is easy to compute the values of $\mathrm{s}(n)$.

The Principle of Mathematical Induction.

Let $p(n)$ *be a condition over* $\mathbb{N}$. *If the following two statements are true, then* $p(n)$ *is true for each integer* $n \in \mathbb{N}$.

1. (Base case) $p(0)$ *is true;*
2. (Inductive step) *For each* $k \in \mathbb{N}$, *if* $p(k)$ *is true, then* $p(k+1)$ *is true.*

Theorem 1.27. *For every* $n \in \mathbb{N}$

$$\sum_{i=0}^{n} 2^i = 2^{n+1} - 1$$

Proof.

The Principle of Strong Mathematical Induction.

Let $p(n)$ *be a condition over* $\mathbb{N}$. *If the following two statements are true, then* $p(n)$ *is true for each integer* $n \in \mathbb{N}$.

1. (Base case) $p(0)$ *is true;*
2. (Inductive step) *For each* $k \in \mathbb{N}$, *if each of* $p(0), p(1), \ldots p(k)$ *is true, then* $p(k+1)$ *is true.*

Recall that a prime number is an integer $n \geq 2$ such that the only positive divisors of n are 1 and itself.

Any integer $n \geq 2$ that is not prime is called composite. For any composite number n there exists integers $2 \leq a, b < n$ such that $\mathrm{n} = \mathrm{a}b$.

Theorem 1.28. *Every integer* $n \geq 2$ *is prime or can be expressed as a product of prime numbers.*

Proof.

Note that induction can start at any base case.

Suppose we want to prove a condition $p(n)$ holds true for all integers $n \geq 2$.

Then applying the **Principle of Strong Mathematical Induction** to prove the condition $q(n)$ holds true for all integers $n \geq 0$ gives us:

If the following two statements are true, then $p(n)$ *is true for each integer* $n \geq 2$.

1. (Base case) $p(2)$ *is true;*
2. (Inductive step) *For each integer* $k \geq 2$, *if each of* $p(2), p(3), \ldots p(k)$ *is true, then* $p(k+1)$ *is true.*

Theorem 1.29. *Every integer* $n \geq 2$ *is prime or can be uniquely expressed as a product of prime numbers.*

Chapter 2:
Set Theory

2.1 Set Theory Preliminaries

We begin with our common understanding of the meaning of the word set.

Definition. *A <u>set</u> is a collection of unique objects. Objects contained in a set are called <u>elements</u>. When A is a set and x is an element of A we write $x \in A$.*

Example.

The elements of a set are unordered.

Example.

Example. Let S be the set of all things that are not elephants.

Russell's Paradox: *Let P be the set of all sets that are not elements of themselves.*

Question. Is $P \in P$?

Barber's Paradox: *There is a barber who shaves all those, and only those, who do not shave themselves.*

Question. Does the barber shave himself?

Definition. (Vector space axioms)

A <u>*vector space*</u> *is a set* V *together with addition and scalar multiplication so that the following properties hold true:*

1. *For all* $x, y \in V$, $x + y \in V$.
2. *For all* $x \in V$ *and* $a \in \mathbb{R}$, $ax \in V$.
3. $x + y = y + x$ *for all* $x, y \in V$.
4. $(x + y) + z = x + (y + z)$ *for all* $x, y, z \in V$.
5. *There exists an element* $0 \in V$ *so that* $x + 0 = x$ *for all* $x \in V$.
6. *For each* $x \in V$ *there exists* $x' \in V$ *so that* $x + x' = 0$.
7. $1x = x$ *for all* $x \in V$.
8. $a(x + y) = ax + ay$ *for all* $x, y \in V$ *and* $a \in \mathbb{R}$.

We call the elements of a vector space <u>*vectors*</u>.

Definition 2.1. (Set Axioms)

- **Equality:** *Two sets are equal when they have the same elements.*
- **Specification:** *If U is a set and $p(x)$ is a condition on U expressed in first-order logic, then the collection of all elements of U for which $p(x)$ is true is a set.*
- **Pairing:** *If A and B are sets, then there is a set that contains A and B as elements.*
- **Union:** *For any collection of sets C, there exists a set that contains all elements that belong to at least one set in C.*
- **Power Set:** *If A is a set then there is a set that contains all of the subsets of A.*
- **Replacement:** *The range of a function is a set.*
- **Inductive Set:** *There exists a set A so that every $x \in A$ is a set, $\emptyset \in A$, and if $x \in A$ then $x \cup \{x\} \in A$.*
- **Foundation:** *Every non-empty set A contains an element y so that A and y have no common elements.*
- **Axiom of Choice:** *Given a collection of sets, there is a set that has exactly one element from each set.*

Let's look at these axioms more carefully.

Axiom of Equality

Definition 2.2. *Let A and B be sets. We say A and B are* <u>*equal*</u> *when they have the same elements. In other words, we say A and B are equal when the following statement is true:*

$$(\forall x)\ x \in A \Leftrightarrow x \in B$$

When A and B are equal, we write $A = B$.

Definition 2.3. *Let A and B be sets. We say B is a* <u>*subset*</u> *of A when every element of B is also an element of A. In other words, we say B is a subset of A when the following statement is true:*

$$(\forall x)\ x \in B \Rightarrow x \in A$$

When B is a subset of A, we write $B \subseteq A$.

When B is a subset of A, and $A \neq B$, we say B is a proper subset of A and we write $B \subsetneq A$.

Aside. *Some authors use the notation $\subset$ in place of $\subsetneq$. Feel welcome to use either in your submitted work.*

Axiom of Union

Definition 2.4. *Let A and B be sets. The* <u>*union*</u> *of A and B is the set containing all elements of A and all elements of B. We denote the union of A and B as $A \cup B$. That is*

$$A \cup B = \{x | \, (x \in A) \vee (x \in B)\}$$

Axiom of Specification

The Axiom of Specification tells us that we may use conditions to specify membership in a set. That is, if U is a set and $p(x)$ is a condition on U, then the collection of all elements of U for which $p(x)$ is true forms a set. We denote such a set as

$$\{x \in U \mid p(x)\}$$

This method of specifying a set is usually referred to as <u>*set-builder notation*</u>.

Definition 2.5. *Let U be a set and let A and B be subsets of U. The* <u>*complement*</u> *of B in A is the set containing all elements A which are not element of B. We denote the complement of B in A as $A \setminus B$. That is*

$$A \setminus B = \{x \in A \mid x \notin B\}$$

Definition 2.6. *Let A be a set. When A contains no elements, we say A is the* <u>*empty set*</u>*. When A is the empty set, we write $A = \{\,\}$.*

Aside. Some authors use the notation $\emptyset$ to denote the empty set. Feel welcome to use $\emptyset$ in your submitted work.

Definition 2.7. *Let U be a set and let A and B be subsets of U. The* <u>*intersection*</u> *of A and B is the subset of U that contains all elements common to both A and B. We denote the intersection of A and B as $A \cap B$. That is*

$$A \cap B = \{x \in U \mid (x \in A) \wedge (x \in B)\}$$

Definition. *Let A and B be sets. For each $a \in A$ and $b \in B$ we define the notation (a, b) to be the set:*

$$(a, b) = \{a, \{a, b\}\}$$

We call (a, b) an <u>ordered pair</u>.

We often regard (a, b) as having 1st coordinate a and 2nd coordinate b.

Notice that the set (a, b) contains two elements, a and $\{a, b\}$, and that $a \in \{a, b\}$.

Using this, we can show that $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Definition 2.10. *Let A and B be sets. The* <u>*Cartesian product*</u> *of A and B is the set of all ordered pairs whose first entry is in A and whose second entry is in B. We denote the Cartesian product of A and B as $A \times B$. That is*

$$A \times B = \{(a, b) \mid (a \in A) \wedge (b \in B)\}$$

Theorem 2.8. *Let A and B be sets. We have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.*

Proof. Let's show that if $A = B$ then $A \subseteq B$ *and* $B \subseteq A$.

1.

2.

Show $A \subseteq B$.

3.

4.

5.

6.

7.

8.

Show $B \subseteq A$.

9.

10.

11.

12.

13.

14.

Conclusion:

15.

16.

Exercise 2.9. *Show that if* $A \subseteq B$ *and* $B \subseteq A$ *then* $A = B$.

4.1 Functions and Relations

We can use the Cartesian product to formalize the notion of a *relation* between sets.

Example. The University of Melbourne has a database of the students enrolled in various subjects.

Let S be the set of all subjects at the University and let P be the set of all students. Then the University's database can be represented as

Definition 4.1. *Let A, B, and R be sets. We say that R is a* <u>*relation*</u> *between A and B when R is a subset of $A \times B$.*

We can regard functions as a type of relation which assigns to each input an output.

Example. Consider the set

$$A = \{(z, z^2) \mid z \in \mathbb{R}\}$$

Definition 4.2. *Let A and B be sets and f be a relation on A and B.*

We say f is a <u>function</u> from A to B when each element $x \in A$ appears as the first entry of an ordered pair (x, y) for exactly one element $(x, y) \in f$.

When f is a function from A to B, we write $f: A \to B$.

We call A the <u>domain</u> of f and B the <u>codomain</u> of f.

We say that R is a relation between A and B when R is a subset of $A \times B$.

Definition 4.3. *For each $(x, y) \in f$ we call*

- *x a <u>preimage</u> of y and*
- *y the <u>image</u> of x.*

We write $f(x) = y$.

Definition 4.4. *Let* $f: A \to B$ *be a function and* $y \in B$.

The <u>*pre-image*</u> *of* y *is the set* $f^{-1}(b) = \{x \in A \mid f(x) = y\}$.

Definition 4.5. *The* <u>*range*</u> *of* f *is the set* $\mathrm{range}(f) = \{f(x) \in B \mid x \in A\}$.

Example.

2.2 Ordered and Bounded Sets

Take a moment to draw a picture of $\mathbb{Z}$, the set of all integers.

Definition 2.11. *Let S be a set. An* <u>*order*</u> *on S is a relation, denoted $<$, so that the following two statements are true:*

(O1) *for all $x, y \in S$ exactly one of the following is true:*

- $x < y$
- $y < x$
- $x = y$

(O2) *for all $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.*

Additionally, we let $x > y$ denote the statement $y < x$. We let $x \leq y$ and $y \geq x$ both denote the statement $(x < y) \wedge (x = y)$.

Examples.

The function $f(x) = \frac{1}{x}$ is bounded below by 0 on the domain $x > 0$.

The function $f(x) = \frac{1}{x}$ is not bounded above on the domain $x > 0$.

Definition 2.12. *Let S be an ordered set and let A be a subset of S. We say that A is* <u>*bounded below*</u> *in S when there exists $\beta \in S$ so that the following statement is true*

$$(\forall x \in A)\ \ \beta \geq x$$

We say that β is an <u>*upper bound*</u> *for A in S.*

Definition 2.13. *Let S be an ordered set and let A be a subset of S. We say that A is* <u>*bounded above*</u> *in S when there exists $\beta \in S$ so that the following statement is true*

$$(\forall x \in A)\ \ \beta \leq x$$

We say that β is a <u>*lower bound*</u> *for A in S.*

Example. $A = \left\{ \frac{1}{r} \middle| r \in \mathbb{R} \text{ and } r > 0 \right\}$ in $S = \mathbb{R}$

Definition 2.14. *Let S be an ordered set and let $A \subseteq S$.*

Let U_A be the set of all upper bounds of A in S.

We say that $\alpha \in U_A$ is a <u>least upper bound</u> of A in S when

$$(\forall \beta \in U_A) \quad \alpha \leq \beta$$

We also call the least upper bound α the <u>supremum</u> of A in S and write

$$\sup A = \alpha$$

Definition 2.15. *Now let L_A be the set of all lower bounds of A in S.*

We say that $\alpha \in L_A$ is a <u>greatest lower bound</u> of A in S when

$$(\forall \beta \in L_A) \quad \alpha \geq \beta$$

We also call the greatest lower bound α the <u>infimum</u> of A in S and write

$$\inf A = \alpha$$

Example 2.16. $A = \{1,2,3,4,10\}$ in $S = \mathbb{N}$

Example 2.17. $A = \{x \in \mathbb{R} \mid x^2 \leq 2\} \cup \{x \in \mathbb{R} \mid x < 0\}$ in $S = \mathbb{R}$

Example 2.17. $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\} \cup \{x \in \mathbb{Q} \mid x < 0\}$ in $S = \mathbb{Q}$

Example suggests:

There is at least one rational number
between every rational and every irrational number.

Let S be an ordered set and let $A \subseteq S$. Let $\alpha \in S$.

Let U_A be the set of all upper bounds of A in S.

To prove $\alpha = \sup A$, we must prove two things:

1. $(\forall x \in A)\ x \leq \alpha$ (α is an upper bound for A)
2. $(\forall \beta \in U_A)\ \alpha \leq \beta$ (α is not bigger than any other upper bound for A)

Let L_A be the set of all lower bounds of A in S.

To prove $\alpha = \inf A$, we must prove two things:

1. $(\forall x \in A)\ \alpha \leq x$ (α is an lower bound for A)
2. $(\forall \beta \in L_A)\ \beta \leq \alpha$ (α is not bigger than any other lower bound for A)

Example. $A = \{x \in \mathbb{R} \mid 1 \leq x < 2\}$

Claim. $\inf A = 1$

First we show that $(\forall x \in A)\ \ x \leq 1$.

1.

2.

Next we show that $(\forall \beta \in L_A)\ \ \beta \leq 1$.

1.

2.

3.

Example. $A = \{x \in \mathbb{R} \mid 1 \leq x < 2\}$

Claim. $\sup A = 2$

First we show that $(\forall x \in A)\ \ x \leq 2$.

1.

2.

3.

Next we show that $(\forall \beta \in U_A)\ \ 2 \leq \beta$.

1.

2.

Example. $A = \{x \in \mathbb{R} \mid 1 \leq x < 2\}$

Claim. $\sup A = 2$

Let's try again… Next we show that $(\forall \beta \in U_A)\ 2 \leq \beta$. We proceed by contraction.

1.
2.
3.
4.
5.
6.
7.
8.
9.
10.

Theorem 2.19. *Let $A \subseteq \mathbb{R}$ be bounded above and non-empty.*

Let $\gamma \in \mathbb{R}$ be an upper bound of A in $\mathbb{R}$.

$\sup A = \gamma$ *if and only if for every $\epsilon > 0$ there exists $x \in A$ such that $x > \gamma - \epsilon$.*

Completeness Axiom. *Every subset $A \subseteq \mathbb{R}$ that is bounded above in $\mathbb{R}$ has a least upper bound in $\mathbb{R}$.*

3.3 Sets and Operations

Definition 2.20. *Let F be a set equipped with two well-defined binary operations, addition* $(+)$ *and multiplication* $(\times)$. *We say that F is a* <u>*field*</u> *when the following statements are true:*

Moreover, the following statements are true:

(A1)	$(\forall x, y \in F)\ \ x + y \in F$	(*addition is closed*)
(A2)	$(\forall x, y \in F)\ \ x + y = y + x$	(*addition is commutative*)
(A3)	$(\forall x, y, z \in F)\ \ (x + y) + z = x + (y + z)$	(*addition is associative*)
(A4)	$(\exists 0 \in F)\ \ [(\forall x \in F)\ \ x + 0 = x]$	(*additive identity*)
(A5)	$(\forall x \in F)\ \ [(\exists y \in F)\ \ x + y = 0]$	(*additive inverse*)
(M1)	$(\forall x, y \in F)\ \ xy \in F$	(*multiplication is closed*)
(M2)	$(\forall x, y \in F)\ \ xy = yx$	(*mult. is commutative*)
(M3)	$(\forall x, y, z \in F)\ \ (xy)z = x(yz)$	(*mult. is associative*)
(M4)	$(\exists 1 \in F \setminus \{0\})\ \ [(\forall x \in F)\ \ 1x = x]$	(*multiplicative identity*)
(M5)	$(\forall x \in F \setminus \{0\})\ \ [(\exists y \in F \setminus \{0\})\ \ xy = 1]$	(*multiplicative inverse*)
(D)	$(\forall x, y, z \in F)\ \ x(y + z) = xy + xz$	(*distributive property*)

Note. The assumption that addition (and multiplication) is a <u>well-defined binary operation</u> means that addition is a function

$$\text{Sum} : F \times F \to F$$

$$\text{Sum}(x, y) = x + y \qquad \text{for each } x, y \in F.$$

In particular, for each $x, y, z \in F$

$$\begin{aligned} x = y \quad &\Longrightarrow \quad \text{Sum}(x, z) = \text{Sum}(y, z) \\ &\Longrightarrow \qquad\quad x + z = y + z \end{aligned}$$

Definition 2.21. *Let F be a field.*

We call 0 *the* <u>*additive identity*</u> *of* F.

Given $x, y \in F$ *such that* $x + y = 0$, *we call* y *the* <u>*additive inverse*</u> *of* x. *When* y *is the additive inverse of* x, *we denote* y *as* $-x$.

Definition 2.22. *We call* 1 *the* <u>*multiplicative identity*</u> *of* F.

Given $x, y \in F$ *such that* $xy = 1$, *we call* y *the* <u>*multiplicative inverse*</u> *of* x. *When* y *is the* <u>*multiplicative*</u> *inverse of* x, *we denote* y *as* $1/x$ *or* x^{-1}.

Example. $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{F}_2 = \{0,1\}$ are well-known fields.

Example. $\mathbb{Z}$ has addition and multiplication operators, but $\mathbb{Z}$ is *not* a field since it does not contain the multiplicative inverses of every non-zero integer (as in (M5)).

Example. Let

$$M = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \middle| \, a, b \in \mathbb{R} \right\}$$

Our experience in Linear Algebra tells us how to add and multiply elements of M.

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} + \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} a+c & -(b+d) \\ b+d & a+c \end{bmatrix}$$

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} c & -d \\ d & c \end{bmatrix} = \begin{bmatrix} ac-bd & -(ad+bc) \\ ad+bc & ac-bd \end{bmatrix}$$

One can verify that M is a field.

Clearly M is closed under addition and multiplication.

The additive identity 0 in M is the 2×2 zero matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

The multiplicative identity 1 in M is the 2×2 identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Theorem 2.23. *Let F be a field and let $x, y, z \in F$. If $x + y = x + z$, then $y = z$.*

Idea of proof.

Proof.

Theorem 2.24. *Let F be a field and let $x, y, z \in F$ with $x \neq 0$. If $xy = xz$, then $y = z$.*

Consider the sequence of subsets

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

The natural numbers $\mathbb{N}$ is informally defined by

One can formally define $\mathbb{N}$ using the Peano Axioms to specify the properties of the successor function $S(n) = n + 1$.

Using Set Theory, one can represent $\mathbb{N}$ by defining $S(A) = A \cup \{A\}$ for each set A and writing

Consider the sequence of subsets

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

The natural numbers $\mathbb{N}$ is not a field because it does not contain additive or multiplicative inverses.

The integers $\mathbb{Z}$ is defined as the set containing every natural number n and its additive inverse $-n$.

$\mathbb{Z}$ is still not a field because it does not contain multiplicative inverses.

The rational numbers $\mathbb{Q}$ is defined as the set of all ratios a/b where $a, b \in \mathbb{Z}$ and $b \neq 0$. We require that

$$\frac{a}{b} = \frac{c}{d} \quad \text{if and only if} \quad ad = bc.$$

$\mathbb{Q}$ is a field.

Theorem. *The real numbers* $\mathbb{R}$

- *are an ordered field,*
- *contain* $\mathbb{Q}$ *as a subfield, and*
- *have the property that every non-empty subset* $A \subseteq \mathbb{R}$ *that is not bounded above in* $\mathbb{R}$ *has a supremum in* $\mathbb{R}$.

Chapter 3:
The Real Numbers

Definition 3.1. *The* <u>*real numbers*</u> $\mathbb{R}$ *is an ordered set equipped with two binary operations, addition* $(+)$ *and multiplication* $(\times)$ *such that the following statements are true:*

(O1) *for every* $x, y \in \mathbb{R}$ *exactly one of the following is true:*

- $x < y$
- $y < x$
- $x = y$

(O2) $(\forall x, y, z \in \mathbb{R})\ \ [(x < y) \wedge (y < z)] \Rightarrow x < z$

(A1) $(\forall x, y \in \mathbb{R})\ \ x + y \in \mathbb{R}$

(A2) $(\forall x, y \in \mathbb{R})\ \ x + y = y + x$

(A3) $(\forall x, y, z \in \mathbb{R})\ \ (x + y) + z = x + (y + z)$

(A4) $(\exists 0 \in \mathbb{R})\ \ [(\forall x \in \mathbb{R})\ \ x + 0 = x]$

(A5) $(\forall x \in \mathbb{R})\ \ [(\exists y \in \mathbb{R})\ \ x + y = 0]$

(M1) $(\forall x, y \in \mathbb{R})\ \ xy \in \mathbb{R}$

(M2) $(\forall x, y \in \mathbb{R})\ \ xy = yx$

(M3) $(\forall x, y, z \in \mathbb{R})\ \ (xy)z = x(yz)$

(M4) $(\exists 1 \in \mathbb{R} \setminus \{0\})\ [(\forall x \in \mathbb{R})\ \ 1x = x]$

(M5) $(\forall x \in \mathbb{R} \setminus \{0\})\ \ [(\exists y \in \mathbb{R} \setminus \{0\})\ \ xy = 1]$

(D) $(\forall x, y, z \in \mathbb{R})\ \ x(y + z) = xy + xz$

(OA) $(\forall x, y, c \in \mathbb{R})\ \ (x < y) \Rightarrow (x + c < y + c)$

(OM) $(\forall x, y \in \mathbb{R})\ \ [(0 < x) \wedge (0 < y)] \Rightarrow (0 < xy)$

(C) *every non-empty subset of* $\mathbb{R}$ *that is bounded above in* $\mathbb{R}$ *has a least upper bound in* $\mathbb{R}$

The Axioms of the Real Numbers assert that

- $\mathbb{R}$ is an ordered field;
- $\mathbb{R}$ is complete, i.e. every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$.

Notice that $\mathbb{Q}$ is an ordered field but is not complete.

It follows from the Axioms that $\mathbb{R}$ contains $\mathbb{Q}$.

We say $\mathbb{Q}$ is a <u>subfield</u> of $\mathbb{R}$ since $\mathbb{Q} \subset \mathbb{R}$ and $\mathbb{Q}$ is an ordered field with the same order $<$ and operations $+$ and $\times$ as $\mathbb{R}$.

In particular, by (A4) and (M4), $\mathbb{R}$ contains 0 and 1. By the Axioms, $1 > 0$.

By successively adding 1 and using (OA), $\mathbb{N} \subset \mathbb{R}$:

Using (A5) and (M5), it follows that $\mathbb{Z} \subset \mathbb{R}$ and $\mathbb{Q} \subset \mathbb{R}$.

One can show that:

- Using the rational numbers $\mathbb{Q}$, one can construct an ordered field **R** which satisfies the Axioms of the Real Numbers.

  (See the Lecture Notes on Dedekind cuts and Cauchy completion.)

- The ordered field that satisfies the Axioms of the Real Numbers is "unique." That is, every ordered fields satisfying the Axioms "looks the same", or is isomorphic.

Theorem 3.8. *Let* $x, y, z \in \mathbb{R}$ *with* $z > 0$. *If* $x < y$, *then* $xz < yz$.

Idea of Proof.

Proof.

Theorem 3.12. *For all* $x, y \in \mathbb{R}$, *if* $x < y$ *then* $(-y) < (-x)$.

Idea of Proof.

Proof.

Theorem. *Let* 0 *be the additive identity in* $\mathbb{R}$ *and* -0 *be its additive inverse in* $\mathbb{R}$. *Then*

$$-0 = 0.$$

Proof. Since 0 is the additive identity and addition is commutative

$$0 + (-0) = (-0) + 0 = (-0).$$

On the other hand, since -0 is the additive inverse of 0

$$0 + (-0) = 0.$$

Therefore, $-0 = 0 + (-0) = 0$.

Theorem 3.13. *For all* $x \in \mathbb{R}$, *if* $x > 0$ *then* $-x < 0$.

Proof. By Theorem 3.12

$$(-x) < (-0).$$

But $-0 = 0$, so in fact

$$(-x) < 0.$$

We often write

$$a < b < c$$

as a shorthand for

$$a < b \quad \text{and} \quad b < c.$$

Theorem 3.14. *Let $x, y \in \mathbb{R}$. If $x < y$, then*

$$x < \frac{x+y}{2} \quad \textit{and} \quad \frac{x+y}{2} < y.$$

Idea of Proof.

Theorem 3.14. *Let* $x, y \in \mathbb{R}$. *If* $x < y$, *then*

$$x < \frac{x+y}{2} \quad \textit{and} \quad \frac{x+y}{2} < y.$$

Proof. Let's first show that if $x < y$ then $x < \frac{x+y}{2}$.

By similar reasoning, if $x < y$ then $\frac{x+y}{2} < y$.

Definition 3.15. Let $a, b \in \mathbb{R}$ with $a < b$.

The <u>closed interval</u> from a to b is the set

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

Definition 3.16. Let $a, b \in \mathbb{R}$ with $a < b$.

The <u>open interval</u> from a to b is the set

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$$

Due to the flexibility of our notation of closed and open intervals, we can represent intervals that are neither closed nor open as follows:

$$
\begin{aligned}
(a, b] &= \{x \in \mathbb{R} \mid a < x \leq b\} \\
[a, b) &= \{x \in \mathbb{R} \mid a \leq x < b\} \\
(a, \infty) &= \{x \in \mathbb{R} \mid a < x\} \\
[a, \infty) &= \{x \in \mathbb{R} \mid a \leq x\} \\
(-\infty, b) &= \{x \in \mathbb{R} \mid x < b\} \\
(-\infty, b] &= \{x \in \mathbb{R} \mid x \leq b\}
\end{aligned}
$$

Definition 3.17. *Let* $x \in \mathbb{R}$. *We define* $|x|$ *so that*

$$|x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Definition. *Let* $x, y \in \mathbb{R}$. *The notation* $|x - y|$ *refers to the distance between* x *and* y *on the real line. When* $y = 0$ *we write* $|x|$ *in place of* $|x - 0|$.

Examples:

$$|x - 5| < 2$$

$$|x + 3| < 2$$

$$|x| \leq a$$

Theorem 3.18. *Let $x, a \in \mathbb{R}$ with $a > 0$. We have that*

$$|x| \leq a \quad \text{if and only if} \quad -a \leq x \text{ and } x \leq a$$

Theorem 3.19. (The Triangle Inequality) *For every $x, y \in \mathbb{R}$ we have that*

$$|x + y| \leq |x| + |y|.$$

Theorem. *For every* $x \in \mathbb{R}$ *we have that*

$$x \leq |x|.$$

Proof. By (O1), we can separately consider the cases $x \geq 0$ and $x < 0$.

Case $x \geq 0$

Case $x < 0$

Theorem. *For every* $x \in \mathbb{R}$ *we have that* $-x \leq |x|$.

Theorem 3.19. (The Triangle Inequality) *For every* $x, y \in \mathbb{R}$ *we have that*

$$|x + y| \leq |x| + |y|.$$

Proof. By (O1), we can separately consider the cases $x + y \geq 0$ and $x + y < 0$.

Case $x + y \geq 0$

Case $x + y < 0$

Theorem 3.20. *For every $x, y \in \mathbb{R}, x < y$, there exists a rational number $r \in \mathbb{Q}$ such that*

$$x < r < y.$$

Theorem 3.21. (Archimedean Principle v1) *The set of natural numbers* $\mathbb{N}$ *is unbounded above in* $\mathbb{R}$.

The following statements are equivalent:

Theorem 3.21. (Archimedean Principle v1) *The set of natural numbers* $\mathbb{N}$ *is unbounded above in* $\mathbb{R}$.

Theorem 3.22. (Archimedean Principle v2) *For every* $x \in \mathbb{R}$ *with* $x > 0$ *there exists* $n \in \mathbb{N}$ *such that* $n - 1 \leq x \leq n$.

Theorem 3.23. (Archimedean Principle v3) *If* $x, y \in \mathbb{R}$ *with* $x, y > 0$ *there exists* $n \in \mathbb{N}$ *such that* $y < nx$.

$$\begin{tikzcd}
 & \textbf{Archimedean Principle v1} \arrow[dl, Leftrightarrow] \arrow[dr, Leftrightarrow] & \\
\textbf{Archimedean Principle v2} \arrow[rr, Leftrightarrow] & & \textbf{Archimedean Principle v3}
\end{tikzcd}$$

Theorem 3.22. (Archimedean Principle v2) *For every* $x \in \mathbb{R}$ *with* $x > 0$ *there exists* $n \in \mathbb{N}$ *such that* $n - 1 \leq x \leq n$.

Proof. By Archimedean Principle v1, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. Hence x is not an upper bound of $\mathbb{N}$. Thus the set

$$\{m \in \mathbb{N} \mid m \geq x\}$$

is non-empty. Choose n to be the smallest element of this set.

Theorem 3.23. (Archimedean Principle v3) *If* $x, y \in \mathbb{R}$ *with* $x, y > 0$ *there exists* $n \in \mathbb{N}$ *such that* $y < nx$.

Proof. Set $t = y/x$.

By Archimedean Principle v1, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. Hence t is not an upper bound of $\mathbb{N}$. Therefore, there exists $n \in \mathbb{N}$ such that $t = y/x < n$.

Theorem 3.20. *For every $x, y \in \mathbb{R}, x < y$, there exists a rational number $r \in \mathbb{Q}$ such that*

$$x < r < y.$$

Proof.

Chapter 4:
Sequences

4.2 Sequences

Definition 4.6. *A* <u>*sequence*</u> f *is a function* $f : \mathbb{N}^+ \to \mathbb{R}$.

Note that we often denote a sequence as

$$(f_n) = (\, f_1, \;\; f_2, \;\; f_3, \;\; f_4, \dots \,)$$

rather than

$$f(1), \;\; f(2), \;\; f(3), \;\; f(4), \dots$$

Example. $f: \mathbb{N}^+ \to \mathbb{R}$ defined by $f(n) = n^2 + 1$

Example. We can approximate π by the sequence

$$(f_n) = (\, 3, \;\; 3.1, \;\; 3.14, \;\; 3.141, \;\; 3.1416, \;\; 3.14159, \dots \,)$$

Example. The sequence

$$(-1,\ \ 1,\ \ -1,\ \ 1,\ \ -1,\ \ 1, \ldots)$$

does not converge, nor does

$$(2,\ \ 4,\ \ 6,\ \ 8,\ \ 10,\ \ 12, \ldots)$$

Example. The sequence

$$(f_n) = \left(4 + (-1)^n \frac{2}{\sqrt{n}}\right) = \left(4 - \frac{2}{\sqrt{1}}, 4 + \frac{2}{\sqrt{2}}, 4 - \frac{2}{\sqrt{3}}, 4 + \frac{2}{\sqrt{4}}, 4 - \frac{2}{\sqrt{5}}\right)$$

Converting these terms to rounded decimals

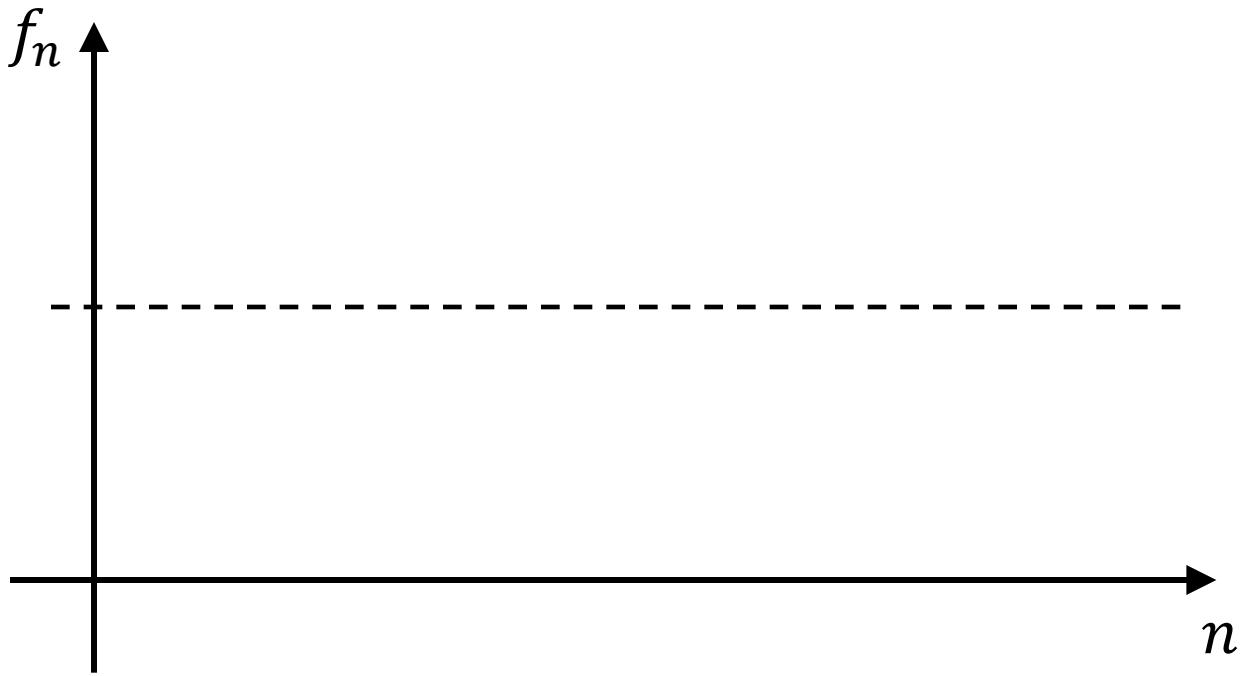
$$(f_n) = (\ 2,\ \ 5.414,\ \ 2.845,\ \ 5,\ \ 3.106,\ \ 4.816,\ \ 3.244,\ \ 4.707, \ldots)$$

Computing more terms

$$f_{100} = 4 + \frac{2}{\sqrt{100}} \approx 4.2 \qquad f_{1000} = 4 + \frac{2}{\sqrt{1000}} \approx 4.0632$$

$$f_{10000} = 4 + \frac{2}{\sqrt{10000}} \approx 4.02 \qquad f_{100000} = 4 + \frac{2}{\sqrt{100000}} \approx 4.0063$$

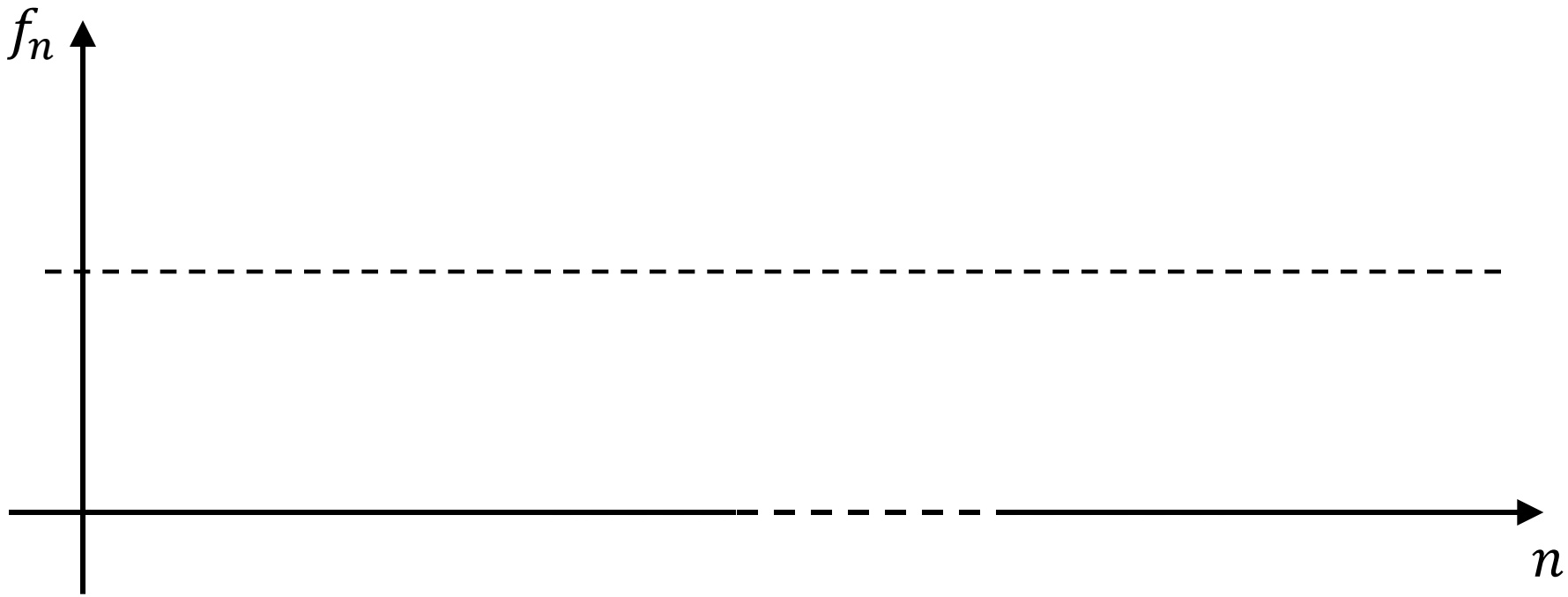
$$f_{100000} = 4 + \frac{2}{\sqrt{100000}} \approx 4.002$$



We expect to say that $\lim_{n\to\infty} f_n = 4$ if

For instance, when $n > 1000$, $|f_n - 4| < 0.0632$,

when $n > 10000$, $|f_n - 4| < 0.02$, etc.



Let $p(n, \epsilon)$ be the condition $|f_n - 4| < \epsilon$.

For example, consider $\epsilon = 0.0001$:

For arbitrary $\epsilon > 0$:

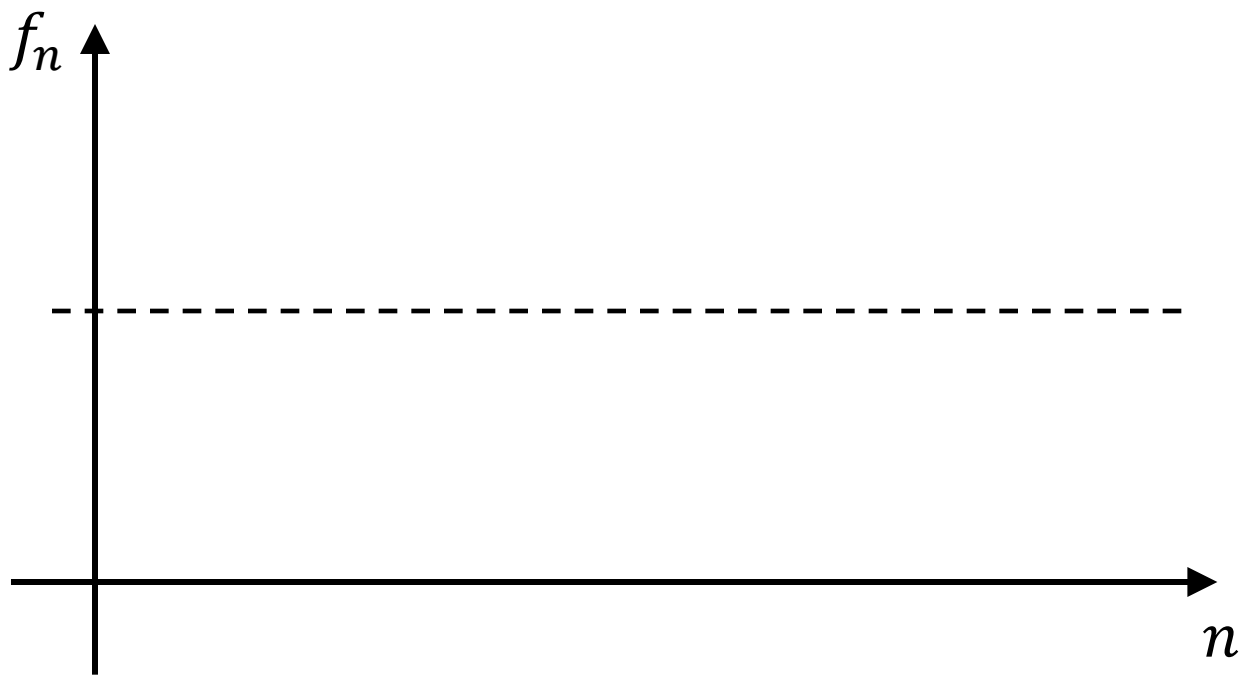
Definition 4.7. *Let* (f_n) *be a sequence and* $L \in \mathbb{R}$.

We say that (f_n) <u>*converges*</u> *to* L *when:*

For every $\epsilon > 0$ *there exists* $M \in \mathbb{N}^+$ *s.t. for all* $n \in \mathbb{N}^+$ *if* $n > M$ *then* $|f_n - L| < \epsilon$.

We write

$$f_n \to L \text{ or } \lim_{n\to\infty} f_n = L$$



Example. Let $f_n = 1 - 1/n$. Let's show that $f_n \to 1$.

Want: $\forall \epsilon > 0 \ \exists M \in \mathbb{N}^+$ s.t. $\forall n \in \mathbb{N}^+$ if $n > M$ then $|f_n - 1| < \epsilon$.

Scrap work. Take any $\epsilon > 0$ and find M:

Proof.

Example. Let $f_n = \frac{1}{n^2+1}$. Let's show that $f_n \to 0$.

Want: $\forall \epsilon > 0 \ \exists M \in \mathbb{N}^+$ s.t. $\forall n \in \mathbb{N}^+$ if $n > M$ then $|f_n - 0| < \epsilon$.

Scrap work. Take any $\epsilon > 0$ and find M:

Proof.

Definition 4.8. *Let* (f_n) *be a sequence. We say* (f_n) <u>*diverges*</u> *when the statement*

$$(f_n) \text{ converges to } L$$

Is false for every $L \in \mathbb{R}$. *In other words, when*

Example. The sequence

$$(-1,\ \ 1,\ \ -1,\ \ 1,\ \ -1,\ \ 1, \ldots)$$

diverges.

Proof.

Theorem 4.9. (Algebra of Limits Theorem)

Let (f_n) *and* (g_n) *be convergent sequence with* $f_n \to \alpha$ *and* $g_n \to \beta$. *Then*

i. $f_n + g_n \to \alpha + \beta$

ii. $f_n - g_n \to \alpha - \beta$

iii. $f_n g_n \to \alpha\beta$

iv. $f_n / g_n \to \alpha / \beta$ (*provided* $g_n \neq 0$ *for all* $n \in \mathbb{N}^+$ *and* $\beta \neq 0$)

Proof of i.

Theorem. *Let* (f_n) *and* (g_n) *be convergent sequences.*

- *If* $f_n \leq g_n$ *for all* $n \in \mathbb{N}^+$*, then*

$$\lim_{n\to\infty} f_n \leq \lim_{n\to\infty} g_n \,.$$

- *If* $f_n \leq b$ *for all* $n \in \mathbb{N}^+$*, then*

$$\lim_{n\to\infty} f_n \leq b.$$

- *If* $a \leq f_n$ *for all* $n \in \mathbb{N}^+$*, then*

$$a \leq \lim_{n\to\infty} f_n \,.$$

Example. Consider the following sequences

i. $(1, 2, 3, 4, 5, \dots)$

ii. $(-1, -2, -3, -4, -5, \dots)$

iii. $(1, 4, 9, 16, 49, \dots)$

iv. $(-2, -4, -8, -16, -32, \dots)$

We expect them to diverge since they are *unbounded*.

That is:

Theorem 4.11. *Let* (f_n) *be a sequence. If* (f_n) *is not bounded, then* (f_n) *does not converge.*

Definition 4.10. Let (f_n) be a sequence. We say that (f_n) is <u>bounded</u> if the set

$$\{ f_n \mid n \in \mathbb{N}^+ \}$$

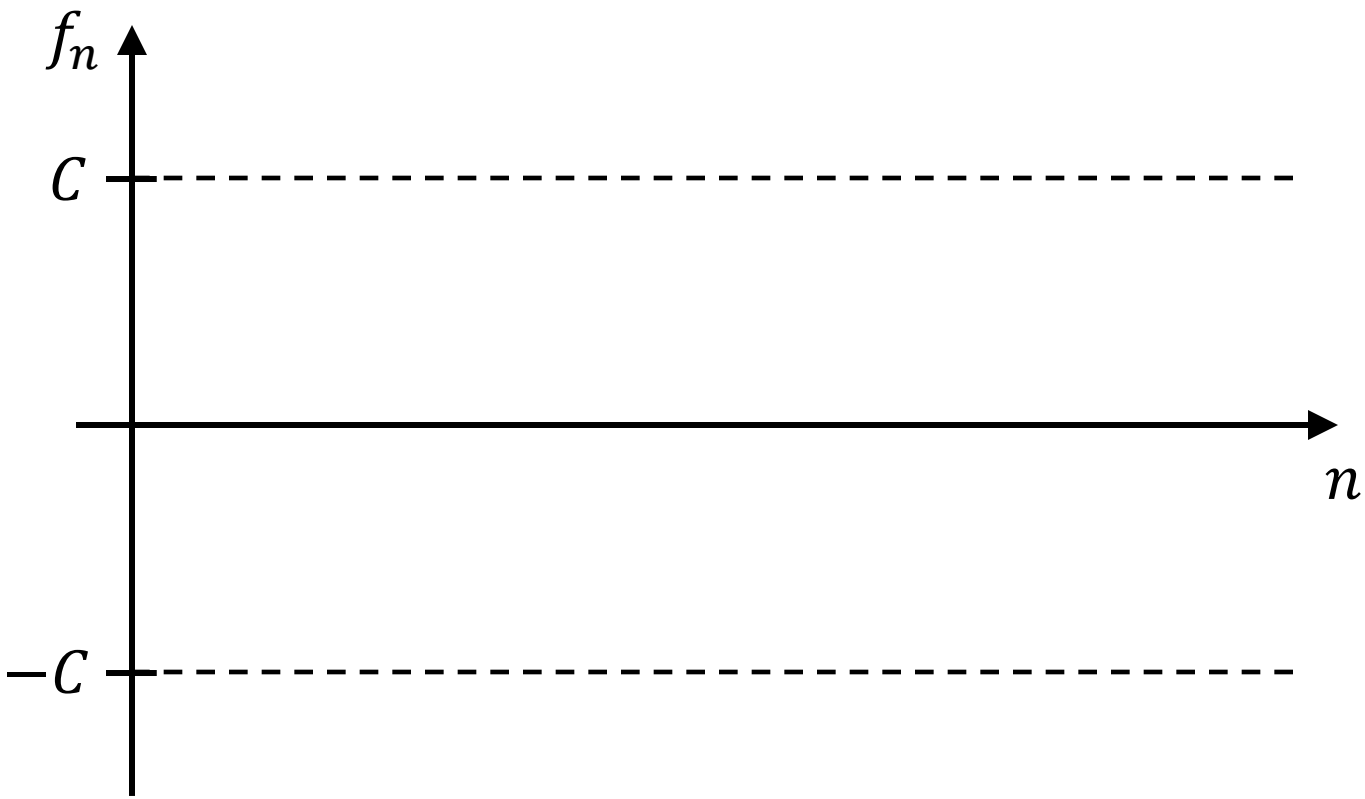
is bounded both above and below.

If (f_n) is not bounded, we say (f_n) is <u>unbounded</u>.

Notice that (f_n) is bounded means that

(f_n) is bounded when

$$-C \leq f_n \leq C \text{ for all } n \in \mathbb{N}^+.$$



Theorem 4.12. *A sequence* (f_n) *is bounded if and only if there exists* $C > 0$ *so that*

$$|f_n| \leq C \text{ for all } n \in \mathbb{N}^+.$$

Now let's state and prove the contrapositive of Theorem 4.11:

Theorem 4.13. *Let* (f_n) *be a sequence. If* (f_n) *converges, then* (f_n) *is bounded.*

Example. The following sequence is bounded but divergent

$$(-1,1,-1,1,\dots)$$

as it is an interleaving of two different sequences

Definition 4.16. *Let* (f_n) *be a sequence and let* $n_1, n_2, n_3, \ldots$ *be integers such that*

$$0 < n_1 < n_2 < n_3 < \cdots$$

We say that

$$(f_{n_1}, f_{n_2}, f_{n_3}, \ldots)$$

is a <u>*subsequence*</u> *of* (f_n). *We denote a subsequence of* (f_n) *as* (f_{n_k}).

Theorem 4.14. *A sequence* (f_n) *converges to* L *if and only if every subsequence of* (f_n) *converges to* L.

Proof.

Theorem 4.14. *A sequence* (f_n) *converges to* L *if and only if every subsequence of* (f_n) *converges to* L.

Notice that Theorem 4.14 contains:

If (f_n) *converges to* L, *then every subsequence of* (f_n) *converges to* L.

The contrapositive of this statement is:

If at least one subsequence of (f_n) *does not converge to* L,
then (f_n) *does not converge to* L.

It follows that:

Corollary 4.15. *Let* (f_n) *be a sequence. If* (f_n) *has two subsequences each of which converge to a different value, then* (f_n) *diverges.*

Corollary 4.17. *Let* (f_n) *be a sequence. If* (f_n) *has a subsequence which diverges, then* (f_n) *diverges.*

Theorem 4.18 (Divergence Criteria Theorem)

Let (f_n) be a sequence. If any of the following are true, then (f_n) diverges:

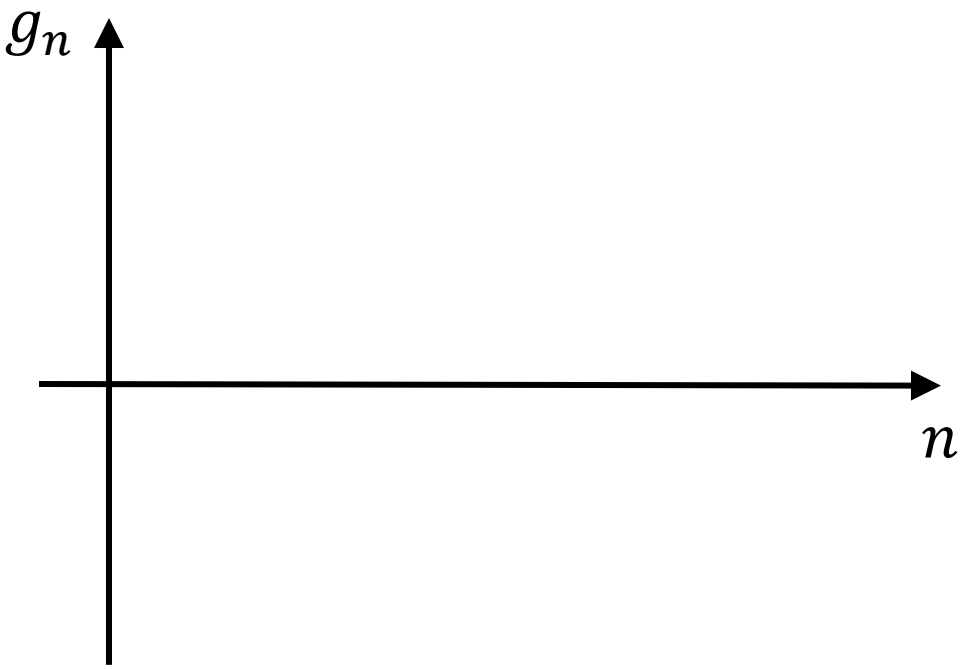
1. (f_n) *is unbounded;*
2. (f_n) *has two subsequences each of which converge to a different value;*
3. (f_n) *has a subsequence that diverges.*

Recall that:

Theorem 4.13. *Let* (f_n) *be a sequence. If* (f_n) *converges, then* (f_n) *is bounded.*

Examples

- $f_n = (-1)^n$ is bounded but divergent
- $g_n = 1 - 1/n$ is bounded and convergent



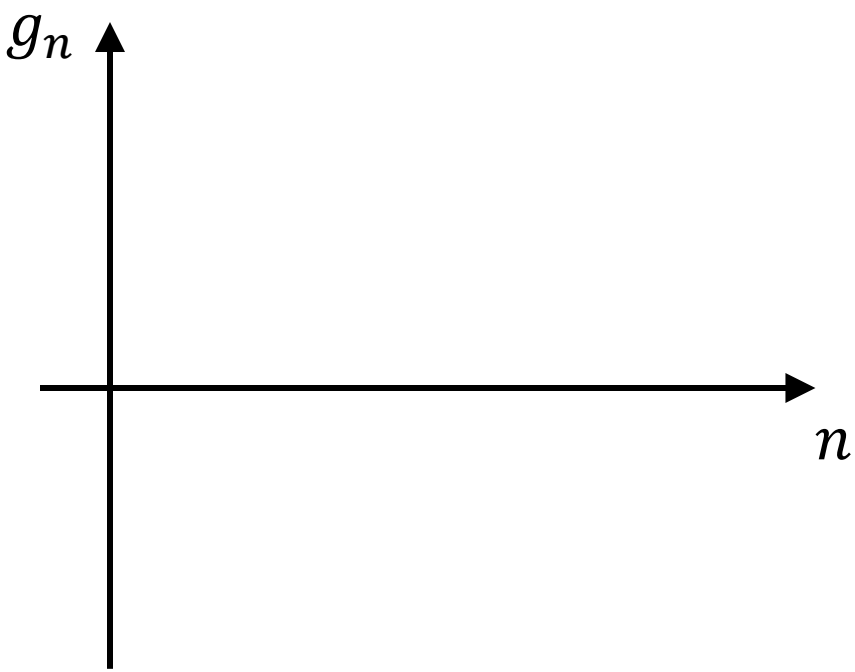
Definition 4.21. *Let (f_n) be a sequence.*

We say that (f_n) is <u>monotone increasing</u> when $f_n \leq f_{n+1}$ for all $n \in \mathbb{N}^+$.

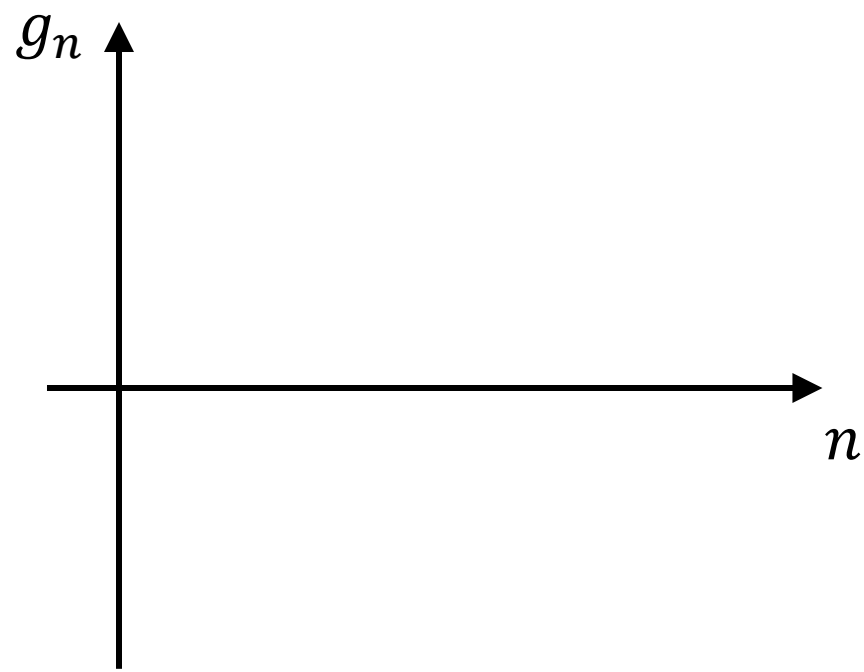
We say that (f_n) is <u>monotone decreasing</u> when $f_n \geq f_{n+1}$ for all $n \in \mathbb{N}^+$.

We say that (f_n) is monotone when (f_n) is either monotone increasing or monotone decreasing.

Increasing



Decreasing



Lemma 4.23. *Let* (f_n) *be a bounded sequence. If* (f_n) *is monotone increasing, then* (f_n) *converges.*

Theorem 4.25. *Let* (f_n) *be a monotone sequence. The sequence* (f_n) *converges if and only if* (f_n) *is bounded.*

Corollary 4.26. *Let* (f_n) *be a bounded sequence and let* $A_f = \{f_n \mid n \in \mathbb{N}^+\}$.

- *If* (f_n) *is monotone increasing, then* $f_n \to \sup A_f$.
- *If* (f_n) *is monotone decreasing, then* $f_n \to \inf A_f$.

Notice that for a bounded increasing monotone sequence (f_n), as n gets larger the terms f_n get squeezed closer and closer to $\sup A_f$.

Hence they get squeezed closer and closer to each other!

In other words:

Definition 4.27. *Let (f_n) be a sequence. We say (f_n) is* <u>*Cauchy*</u> *(or is a* <u>*Cauchy sequence*</u>*) when:*

For every $\epsilon > 0$ there exists $M \in \mathbb{N}^+$ such that if $j, k > M$ then $|f_j - f_k| < \epsilon$.

Example 4.28. Prove $(f_n) = \left(\frac{(-1)^n}{n}\right)$ is Cauchy.

Theorem 4.32 (Cauchy Convergence Criterion)

Let (f_n) *be a sequence. The sequence* (f_n) *converges if and only if* (f_n) *is Cauchy*.

To prove this theorem, we need to prove:

1. If (f_n) converges, then (f_n) is Cauchy.
2. If (f_n) is Cauchy, then (f_n) converges.

Theorem 4.29 *Let* (f_n) *be a sequence. If* (f_n) *converges then* (f_n) *is Cauchy*.

Theorem. *Let* (f_n) *be a sequence. If* (f_n) *is Cauchy then* (f_n) *converges*.

To prove this theorem, we need to prove:

1. Bolzano-Weierstrass Theorem. If (f_n) is a bounded sequence, then (f_n) has a convergent subsequence.

2. If (f_n) is a Cauchy sequence, then (f_n) is bounded.

3. If (f_n) is a Cauchy sequence and (f_n) has at least one convergent subsequence, then (f_n) is convergent.

Theorem. *Let (f_n) be a sequence. If (f_n) is Cauchy then (f_n) converges.*

To prove this theorem, we need to prove:

1. Bolzano-Weierstrass Theorem. If (f_n) is a bounded sequence, then (f_n) has a convergent subsequence.

2. If (f_n) is a Cauchy sequence, then (f_n) is bounded.

   Idea of proof. Argue as in Theorem 4.13 (convergent sequences are bounded) that for $\epsilon = 1$ there exists $M \in \mathbb{N}^+$ such that if $j, k > M$ then $\left|f_j - f_k\right| < 1$.

   Hence if $k > M$ then $|f_k| < |f_{M+1}| + 1$.

3. If (f_n) is a Cauchy sequence and (f_n) has at least one convergent subsequence, then (f_n) is convergent.

Bolzano-Weierstrass Theorem. *If* (f_n) *be a bounded sequence, then* (f_n) *has a convergent subsequence*.

Idea of proof: Show that (f_n) contains a monotone subsequence (f_{n_k}). Since (f_{n_k}) is also bounded, then (f_{n_k}) converges.

Definition. (f_n) *has a* <u>*peak*</u> *at* $n = s$ *if* $n > s \Rightarrow f_n \leq f_s$.

Case 1. (f_n) *has infinitely many peaks at* $n = n_k$ *for* $n_1 < n_2 < n_3 < \cdots$.

Bolzano-Weierstrass Theorem. *If* (f_n) *be a bounded sequence, then* (f_n) *has a convergent subsequence*.

Case 2. (f_n) *has finitely many peaks*.

Theorem. *If* (f_n) *is a Cauchy sequence and* (f_n) *has at least one convergent subsequence, then* (f_n) *converges*.

Definition 4.33. *A sequence* (f_n) *is called contractive if there exists a constant* c *with* $0 < c < 1$ *such that for all* $n \in \mathbb{N}^+$

$$|f_{n+2} - f_{n+1}| \leq c|f_{n+1} - f_n|$$

Theorem 4.34. *If* (f_n) *is a contractive, then* (f_n) *converges.*

Example. Consider the polynomial $f(x) = x^7 - 5x + 1$. From plotting the graph of $y = f(x)$ we can guess that this polynomial has three real roots, and there is a unique root r of $f(x)$ with $0 < r < 1/2$.

4.3 Sequences of Functions

So far, we have only studied sequences of real numbers.

We can also study sequences of other mathematical objects, such as functions.

Example. For all $n \in \mathbb{N}^+$ let $g_n(x)$ denote the following function (of n and x):

$$g_n(x) = \frac{2x^2 + 2nx}{n}$$

For each particular value of n we produce a function $g_n: \mathbb{R} \to \mathbb{R}$.

Example. For all $n \in \mathbb{N}^+$ let $g_n(x)$ denote the following function (of n and x):

$$g_n(x) = \frac{2x^2 + 2nx}{n}$$

For each particular value of x in the domain, we produce a sequence of real numbers $(g_n(x))$.

Definition 4.36. *Let $D \subseteq \mathbb{R}$. Let $(f_n(x))$ be a sequence of functions $f_n : D \to \mathbb{R}$ and let $f : D \to \mathbb{R}$ be a function.*

We say that $(f_n(x))$ <u>converges pointwise</u> to $f(x)$ when

$$\textit{for every } x \in D \textit{ we have } f_n(x) \to f(x).$$

In other words

$$\forall x \in D\ \forall \epsilon > 0\ \exists M = M(x, \epsilon) \in \mathbb{N}^+ \textit{ s.t. if } n > M \textit{ then } |f_n(x) - f(x)| < \epsilon.$$

Now recall our definition of convergence of a sequence (f_n):

Definition. *We say that a sequence* (f_n) <u>*converges*</u> *to* L *when*

$$\forall \epsilon > 0 \; \exists M \in \mathbb{N}^+ \textit{ such that if } n > M \textit{ then } |f_n - L| < \epsilon.$$

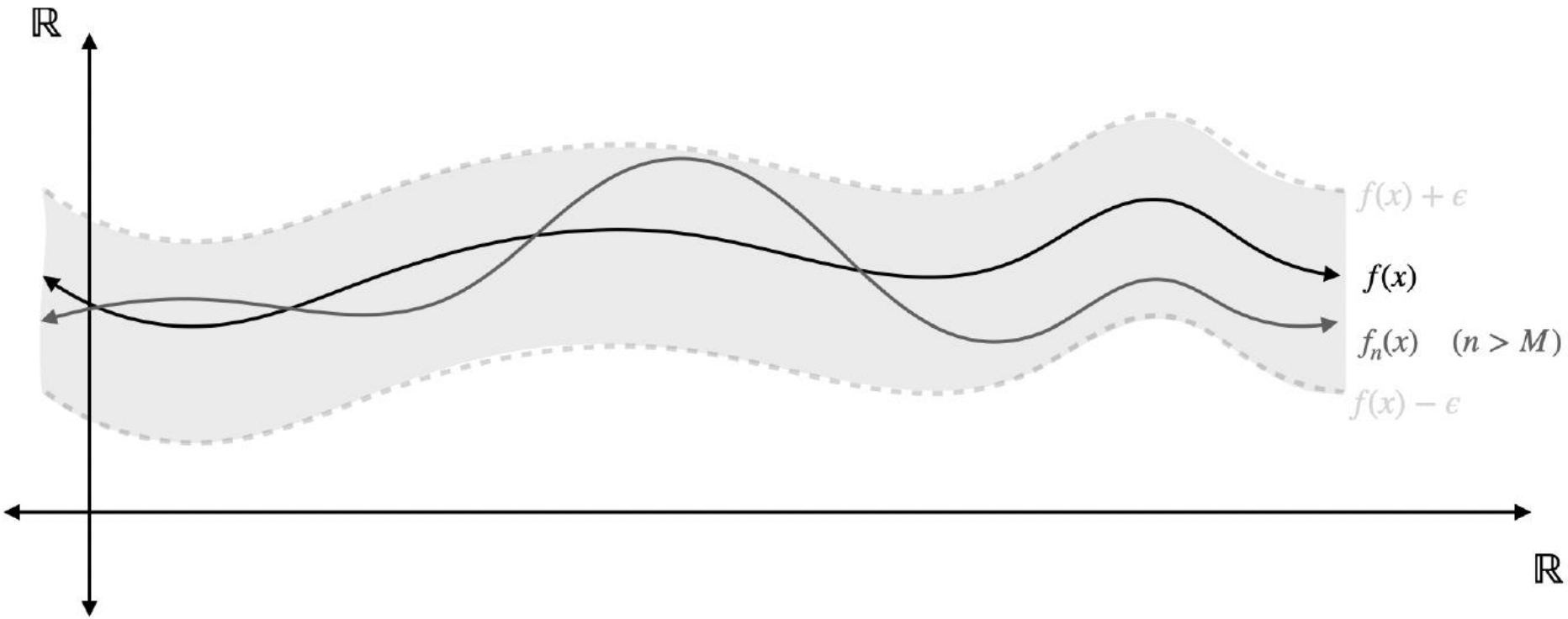
In other words, f_n estimates L with an error $< \epsilon$.

Unfortunately, for sequences of functions, $f_n(x)$ can converge to $f(x)$ at different rates for different values of x.

Definition 4.37. *Let $D \subseteq \mathbb{R}$. Let $(f_n(x))$ be a sequence of functions $f_n : D \to \mathbb{R}$ and let $f : D \to \mathbb{R}$ be a function.*

We say that $(f_n(x))$ <u>converges uniformly</u> to $f(x)$ when

$$\forall \epsilon > 0 \; \exists M = M(\epsilon) \in \mathbb{N}^+ \; s.t. \; \forall x \in D \; if \; n > M \; then \; |f_n(x) - f(x)| < \epsilon.$$



Example. For all $n \in \mathbb{N}^+$ let $g_n(x)$ denote the following function (of n and x):

$$g_n(x) = \frac{2x^2 + 2nx}{n}$$

Show that $(g_n(x))$ does not converge uniformly to $2x$ for $x \in \mathbb{R}$.

Chapter 5:
Limits and Continuity

5.1 Limits

We now want to define the limit of a function. In Calculus you might have learned that

$$\lim_{x \to 3} 2x = 6$$

since

$\boldsymbol{x}$	$\boldsymbol{f(x) = 2x}$
4	8
3.5	7
3.1	6.2
3.01	6.02
3.001	6.002

Example. Define $f: \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

What is

$$\lim_{x \to \sqrt{2}} f(x) = ?$$

$\boldsymbol{x}$	$\boldsymbol{f(x)}$
1	1
1.4	1
1.41	1
1.414	1
1.4142	1

Hence we might suspect

$$\lim_{x \to \sqrt{2}} f(x) = 1$$

Example. Define $f: \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

What is

$$\lim_{x \to \sqrt{2}} f(x) = ?$$

$\boldsymbol{x}$	$\boldsymbol{f(x)}$
$\sqrt{2} - 0.1$	0
$\sqrt{2} - 0.01$	0
$\sqrt{2} - 0.001$	0
$\sqrt{2} - 0.0001$	0
$\sqrt{2} - 0.00001$	0

Depending on which sequence of values we choose for x, we seem to compute that the limit is 0 or 1.

Definition 4.7. *Let (f_n) be a sequence and $L \in \mathbb{R}$.*

We say that (f_n) converges to L when:

For every $\epsilon > 0$ there exists $M \in \mathbb{N}^+$ such that if $n > M$ then $|f_n - L| < \epsilon$.

Conceptually

For every $\epsilon > 0$ if n is close enough to ∞, i.e. n is large, then $|f_n - L| < \epsilon$.

We can similarly describe the limit of a function $f(x)$ as $x \to a$ by

For every $\epsilon > 0$ if x is close enough to a, then $|f(x) - L| < \epsilon$.

By "x is close to a" we mean that for some $\delta > 0$ we have the distance of x to a is $< \delta$, i.e. $|x - a| < \delta$.

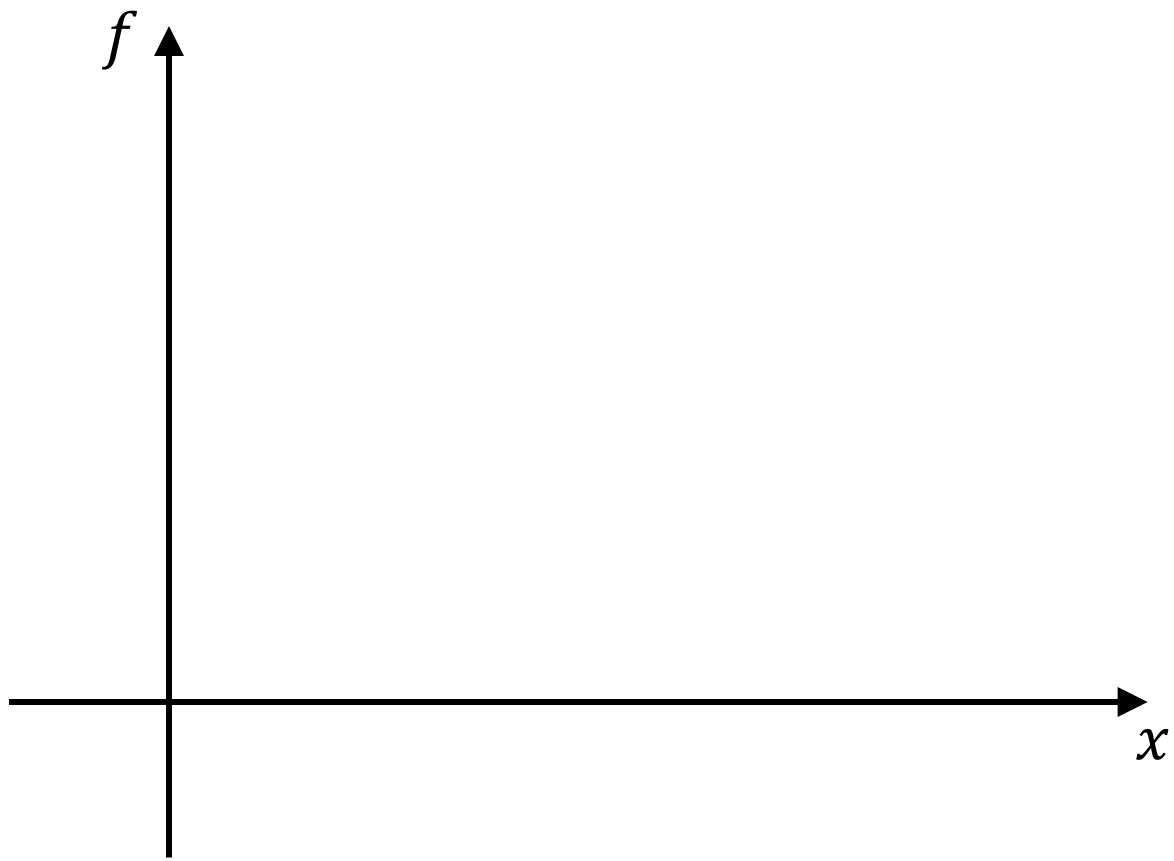
$\longrightarrow$

Proposed Definition. *Let* $f: \mathbb{R} \to \mathbb{R}$ *be function,* $a \in \mathbb{R}$ *and* $L \in \mathbb{R}$. *We write*

$$\lim_{x \to a} f(x) = L$$

when

for all $\epsilon > 0$ *there exists* $\delta > 0$ *such that if* $0 < |x - a| < \delta$ *then* $|f(x) - L| < \epsilon$.



Example. Prove

$$\lim_{x \to 3} 2x = 6$$

Example. Prove

$$\lim_{x \to 3} 2x = 6$$

Scrap work:

Proof:

Example. Consider

$$\lim_{x \to 9} \sqrt{x} = 3$$

The domain of $\sqrt{x}$ is $\{ x \in \mathbb{R} \mid x \geq 0 \}$, but we have only defined limits a functions whose domain is $\mathbb{R}$.

Proposed Definition 2. *Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ be function, $a \in \mathbb{R}$ and $L \in \mathbb{R}$. We write*

$$\lim_{x \to a} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in E$ and $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$.

For which values of a should we consider limits as $x \to a$?

Example. Consider

$$\lim_{x \to -5} \sqrt{x}.$$

This is nonsense since $\sqrt{x}$ is not defined for any x near -5.

Maybe we should require $a \in E$, i.e. a is in the domain of f…

Example. Consider

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}.$$

This limit should make sense but $x = 2$ is not in the domain of $\frac{x^2-4}{x-2}$.

Definition 5.1. *Let $E \subseteq \mathbb{R}$ and $a \in \mathbb{R}$. We say that a is a* <u>*limit point*</u> *of E if for every $\delta > 0$ there exists a point $x \in E$ such that $0 < |x - a| < \delta$.*

Theorem. *Let $E \subseteq \mathbb{R}$ and $a \in \mathbb{R}$. a is a limit point of E if and only if there exists a sequence* (x_n) of points such that $x_n \in E$, $x_n \neq a$, and $x_n \to a$.

Example. Consider the function $f: (0, \infty) \to \mathbb{R}$ defined by $f(x) = \log x$.

The domain of f is $E = (0, \infty)$.

The set of limit points of $E = (0, \infty)$ is

Example. Consider the set $E = \mathbb{N}$ of all natural numbers.

The set of limit points of $E = \mathbb{N}$ is

Official Definition 5.2. *Let $E \subseteq \mathbb{R}$, a be a limit point of E, $f: E \to \mathbb{R}$ be function, and $L \in \mathbb{R}$. We write*

$$\lim_{x \to a} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in E$ and $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$.

Example. Show that

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = 4.$$

Scrap work:

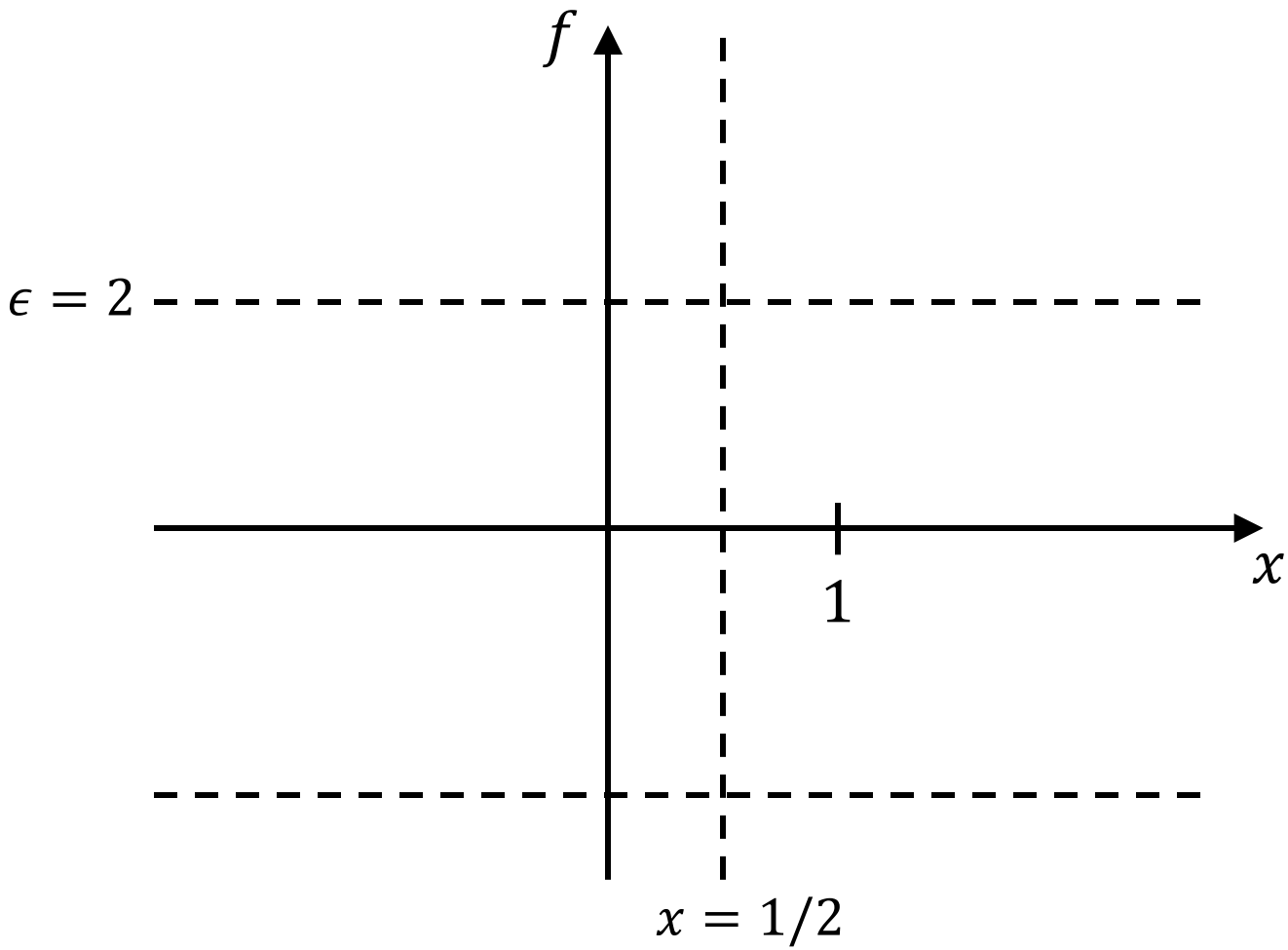
Proof:

Example. Show that

$$\lim_{x \to 1} \frac{x^2 - 1}{2x - 1} = 0.$$

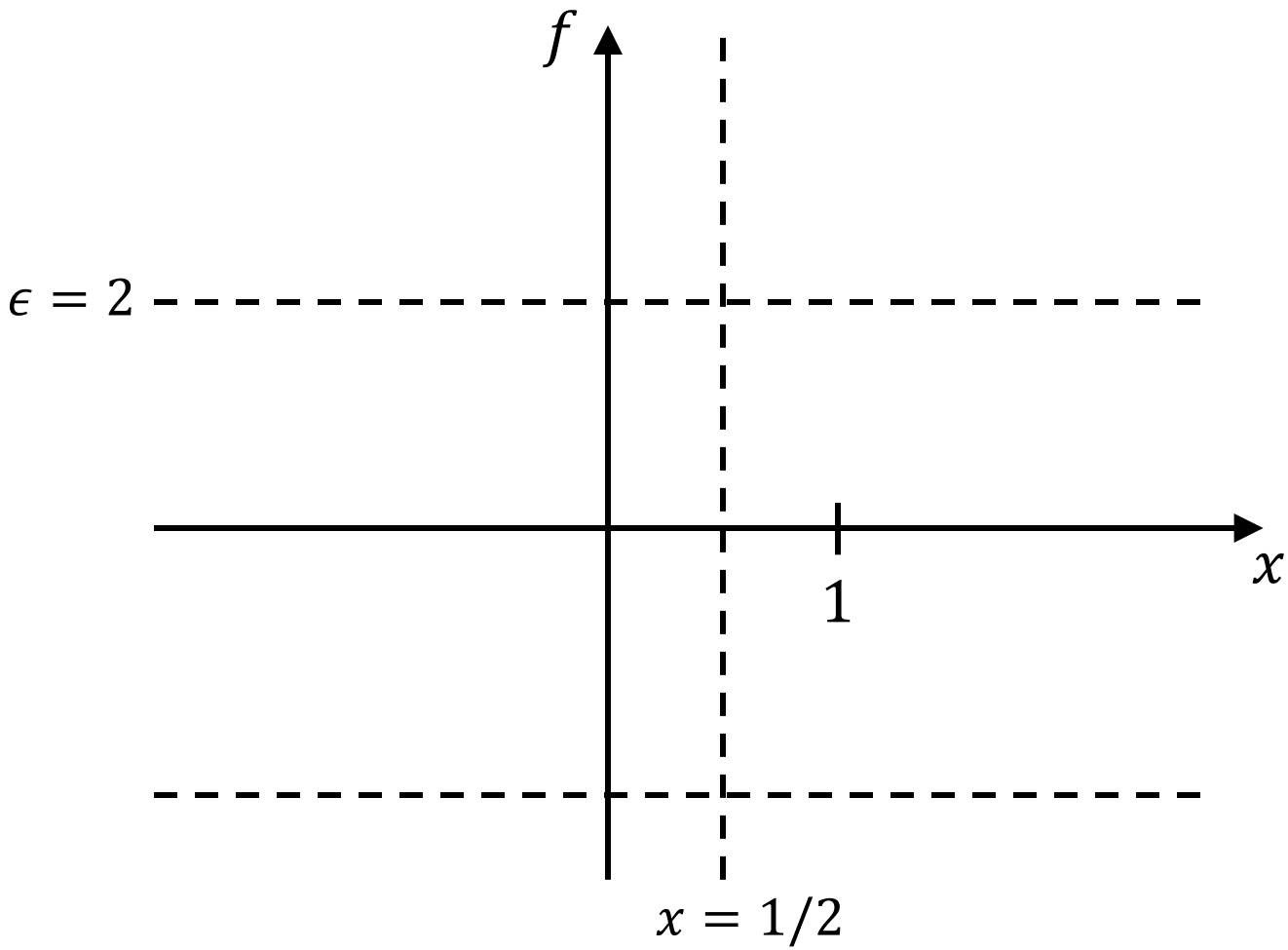
Example. Show that

$$\lim_{x \to 1} \frac{x^2 - 1}{2x - 1} = 0.$$



Example. Show that

$$\lim_{x \to 1} \frac{x^2 - 1}{2x - 1} = 0.$$



Example. Show that

$$\lim_{x \to 1} \frac{x^2 - 1}{2x - 1} = 0.$$

Theorem 5.3. (Algebra of Limits Theorem)

Let $E \subseteq \mathbb{R}$, *let* a *be a limit point of* E, *and* $f, g: E \to \mathbb{R}$. *If*

$$\lim_{x \to a} f(x) = \alpha \qquad \text{and} \qquad \lim_{x \to a} g(x) = \beta$$

then

i. $\lim_{x \to a} (f(x) + g(x)) = \alpha + \beta$;

ii. $\lim_{x \to a} f(x) \cdot g(x) = \alpha \cdot \beta$;

iii. $\lim_{x \to a} f(x)/g(x) = \alpha/\beta$ (*provided* $\beta \neq 0$)

Theorem. *Let $E \subseteq \mathbb{R}$, let a be a limit point of E, and $f, g: E \to \mathbb{R}$. Assume that $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ both exist.*

- *If $f(x) \leq g(x)$ for all $x \in E$, then*

$$\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x).$$

- *If $f(x) \leq c$ for all $x \in E$, then*

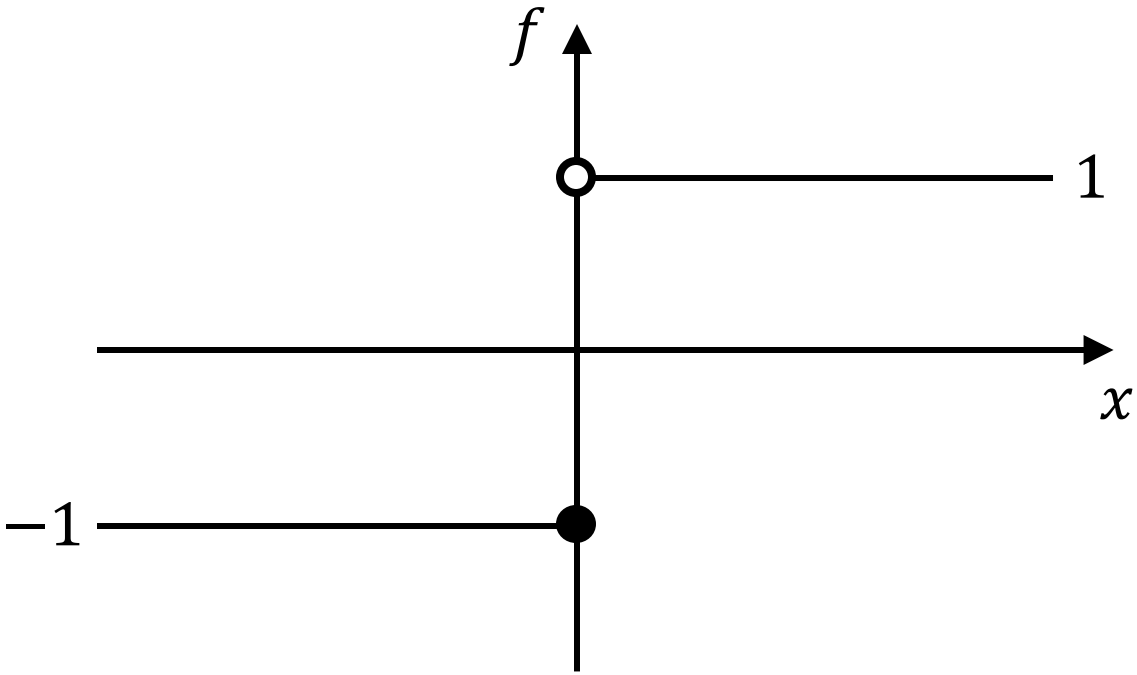
$$\lim_{x \to a} f(x) \leq c.$$

- *If $b \leq f(x)$ for all $x \in E$, then*

$$b \leq \lim_{x \to a} f(x).$$

Example. Define $f: \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x \leq 0 \end{cases}$$



Definition 5.4. *Let $E \subseteq \mathbb{R}$, a be a limit point of E and $f: E \to \mathbb{R}$ be function. We say $f(x)$ <u>diverges</u> as x approaches a if*

$$\lim_{x \to a} f(x) = L$$

is false for all $L \in \mathbb{R}$.

Looking back at our definition of limit, let us focus on:

$$0 < |x - a| < \delta$$

The left-side gives us:

The right-side gives us:

Definition 5.5. *Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ be function and $L \in \mathbb{R}$. Suppose that a is a limit point of $E \cap (a, \infty)$. We write*

$$\lim_{x \to a^+} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in E$ and $a < x < a + \delta$ then $|f(x) - L| < \epsilon$.

Definition 5.6. *Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ be function and $L \in \mathbb{R}$. Suppose that a is a limit point of $E \cap (-\infty, a)$. We write*

$$\lim_{x \to a^-} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in E$ and $a - \delta < x < a$ then $|f(x) - L| < \epsilon$.

Theorem 5.7. *Let* $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ *be function and* $L \in \mathbb{R}$. *Suppose that* a *is a limit point of both* $E \cap (-\infty, a)$ *and* $E \cap (a, \infty)$. *Then*

$$\lim_{x \to a} f(x) = L$$

if and only if

$$\lim_{x \to a^-} f(x) = L \qquad \text{and} \qquad \lim_{x \to a^+} f(x) = L.$$

Definition 5.9. *Let $E \subseteq \mathbb{R}$, a be a limit point of E, $f: E \to \mathbb{R}$ a function. We write*

$$\lim_{x \to a} f(x) = \infty$$

if for every $r \in \mathbb{R}$ there exists $\delta > 0$ such that if $x \in E$ and $0 < |x - a| < \delta$ then $f(x) > r$.

Example. Show that

$$\lim_{x \to 0} \frac{1}{x^2} = \infty$$

Scrap work:

Proof:

Definition 5.10. *Let $E \subseteq \mathbb{R}$ be a set which is not bounded above. Let $f: E \to \mathbb{R}$ be function and $L \in \mathbb{R}$. We write*

$$\lim_{x \to \infty} f(x) = L$$

if for every $\epsilon > 0$ there exists $M \in \mathbb{R}$ s.t. if $x \in E$ and $x > M$ then $|f(x) - L| < \epsilon$.

Definition 5.11. *Let $E \subseteq \mathbb{R}$ be a set which is not bounded above. Let $f: E \to \mathbb{R}$ be function. We say $f(x)$ <u>diverges</u> as x approaches infinity if*

$$\lim_{x \to \infty} f(x) = L$$

is false for all $L \in \mathbb{R}$.

Definition 5.12. *Let $E \subseteq \mathbb{R}$ be a set which is not bounded above. Let $f: E \to \mathbb{R}$ be a function. We write*

$$\lim_{x \to \infty} f(x) = \infty$$

if for every $r \in \mathbb{R}$ there exists $M \in \mathbb{R}$ s.t. if $x \in E$ and $x > M$ then $f(x) > r$.

Example. Show that $\sin x$ diverges as x approaches infinity.

Example. Show that

$$\lim_{x \to \infty} (2x - 1) = \infty$$

5.2 Continuity

Proposed Definition. *Let $E \subseteq \mathbb{R}$ and let $f: E \to \mathbb{R}$. We say that f is continuous when for every $a \in E$*

$$\lim_{x \to a} f(x) = f(a)$$

In terms of our definition of limits, this says:

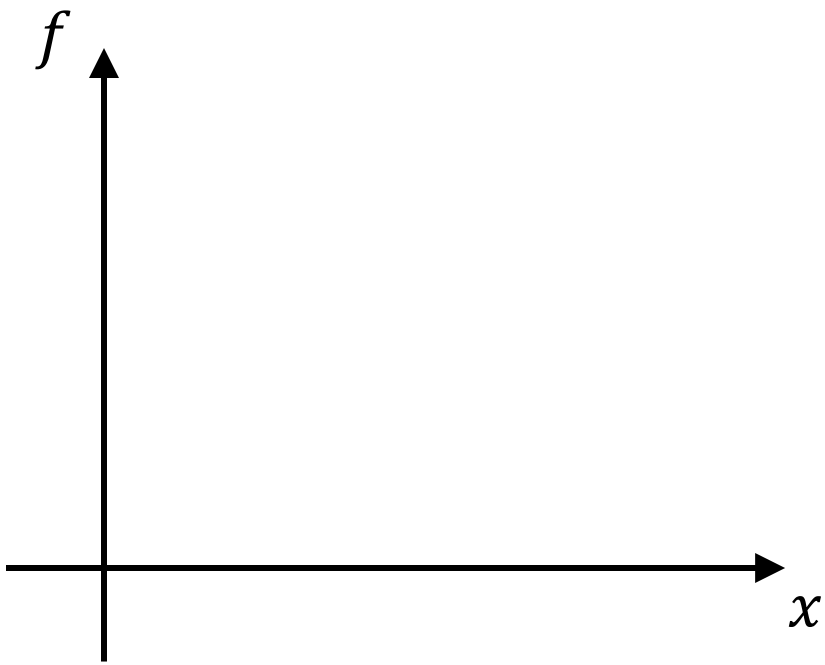
Official Definition 5.13. *Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ a function, and $a \in E$. We say that f is <u>continuous</u> at a if, for every $\epsilon > 0$, there exists $\delta > 0$ such that if $x \in E$ and $|x - a| < \delta$ then $|f(x) - f(a)| < \epsilon$.*

We say f is <u>continuous</u> on E if f is continuous at every point $a \in E$.

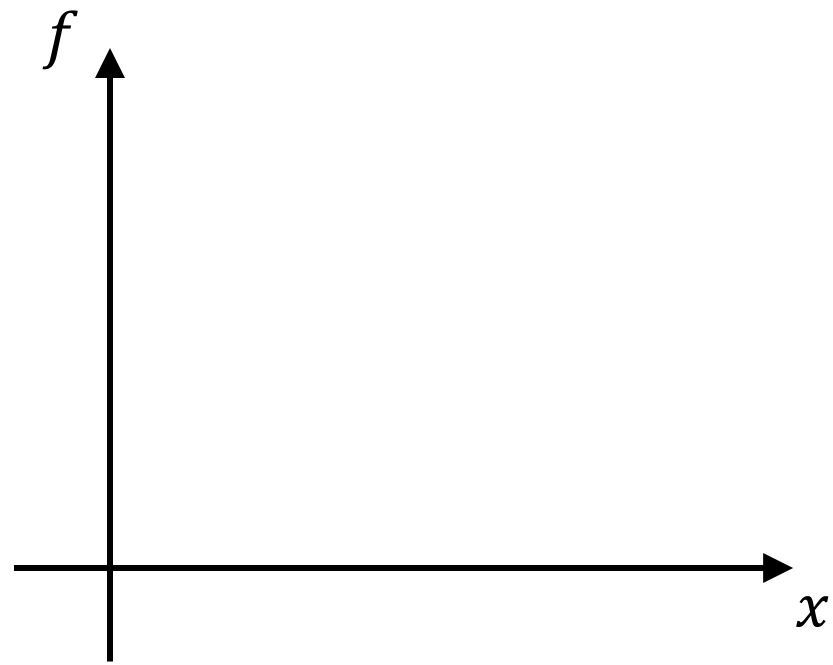
Example. Consider $f(x) = x^2$ at $x = 2$.

$f(x)$ is continuous at $x = 2$ means for any choice of $\epsilon > 0$ we can find $\delta > 0$ such that if $2 - \delta < x < 2 + \delta$ then $4 - \epsilon < x^2 < 4 + \epsilon$. Consider $\epsilon = 1/2$.

Bad choice of δ



Good choice of δ



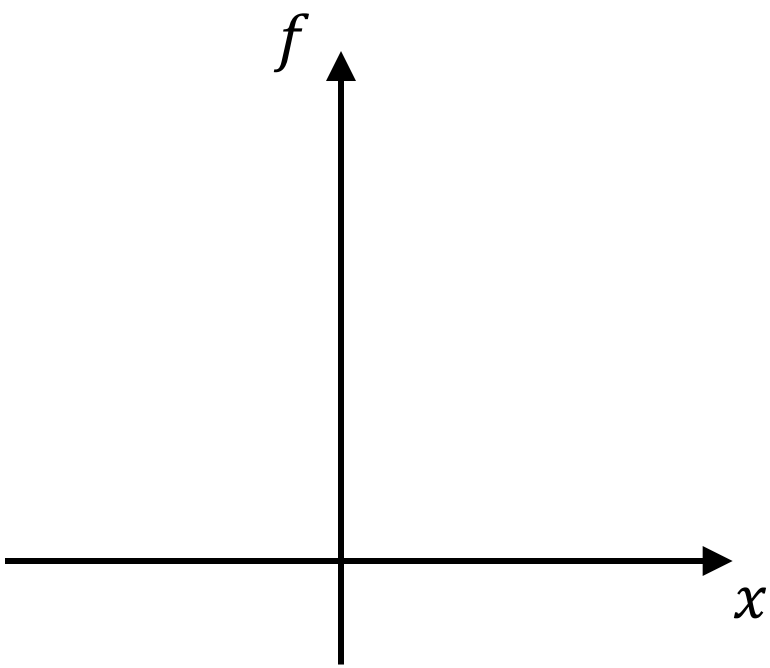
Example 5.14. Show that $f(x) = x$ is continuous on $\mathbb{R}$.

Example 5.15. Show that $f(x) = 1$ is continuous on $\mathbb{R}$.

Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ a function, and $a \in E$. f is not continuous at a when

Example 5.16. Show that $f(x)$ is not continuous at $x = 0$ where

$$f(x) = \begin{cases} 2x + 1 & \text{if } x \neq 0 \\ 5 & \text{if } x = 0 \end{cases}$$



Theorem 5.17. *Let $E \subseteq \mathbb{R}$, and $a \in E$ be a limit point of E. Let $f: E \to \mathbb{R}$. The function f is continuous at a if and only if*

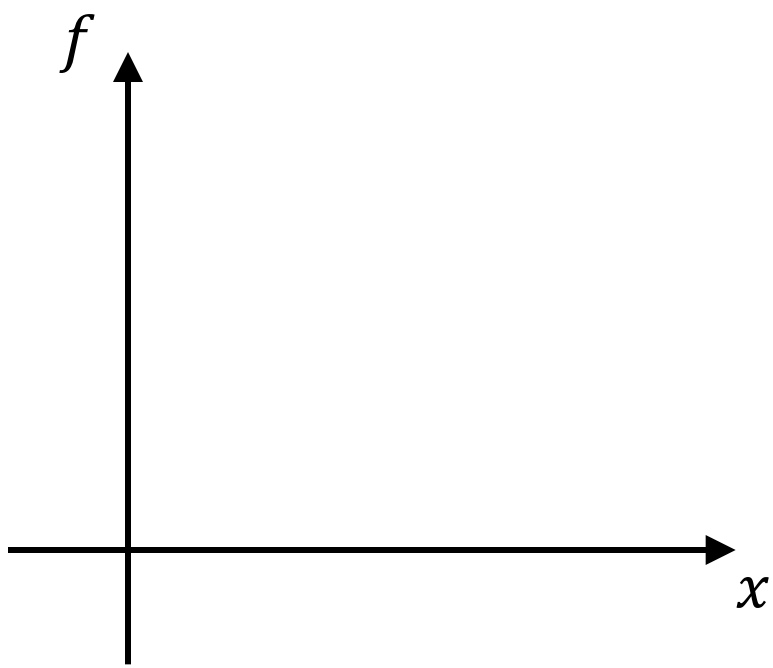
$$\lim_{x \to a} f(x) = f(a).$$

Aside. Our definition of continuity at a does <u>*not*</u> require a to be a limit point of E.

If a is not a limit point of E, then for $\delta > 0$ small, the only $x \in E$ with $|x - a| < \delta$ is $x = a$. Hence $|f(x) - f(a)| = 0 < \epsilon$.

Therefore:

If a is not a limit point of E, then f must be continuous at a.



Theorem 5.18. (Algebra of Continuous Functions)

Let $E \subseteq \mathbb{R}$, $f, g: E \to \mathbb{R}$, *and* $a \in E$. *If* $f(x)$ *and* $g(x)$ *are both continuous at* a, *then*

1. $k\, f(x)$ *is continuous at* a *for all* $k \in \mathbb{R}$;
2. $f(x) + g(x)$ *is continuous at* a;
3. $f(x) \cdot g(x)$ *is continuous at* a;
4. $f(x)/g(x)$ *is continuous at* a *provided* $g(a) \neq 0$.

Corollary 5.20. *For every $n \in \mathbb{N}$, every polynomial of degree n*

$$h(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n$$

is continuous on $\mathbb{R}$.

Theorem 5.21. *The following functions are continuous:*

1. $e^x, \cos(x), \sin(x)$ *are continuous on* $\mathbb{R}$;
2. $\log(x)$ *is continuous on* $(0, \infty)$;
3. $x^{1/p}$ *is continuous on* $(0, \infty)$ *for each* $p \in \mathbb{N}^+$.
4. $c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n$ *is continuous on* $\mathbb{R}$.

Note. For this theorem to make sense, we need to define $e^x, \cos(x), \sin(x)$ etc. One approach involves power series, which we will discuss later in the course.

Theorem. *Let* $A, B \subseteq \mathbb{R}$. *Let* $f: A \to \mathbb{R}$ *and* $g: B \to \mathbb{R}$ *be functions s.t* $f(A) \subseteq B$. *If* $a \in A$ *such that* $f(x)$ *is continuous at* a *and* $g(x)$ *be continuous at* $f(a)$, *then* $g(f(x))$ *is continuous at* a.

Now we want to prove the following facts about a continuous function $f: [a, b] \to \mathbb{R}$.

Theorem 5.25. *If f is continuous on $[a, b]$, then f is bounded on $[a, b]$.*

Theorem 5.28 (Extreme Value Theorem).
If f is continuous on $[a, b]$, then there exists $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \text{ for all } x \in [a, b].$$

Theorem 5.31. (Intermediate Value Theorem)
If f is continuous on $[a, b]$ and $f(a) < y < f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.

Definition 5.22 and 5.23. *Let* $a \in \mathbb{R}$ *and* $\delta > 0$.

The <u>*open*</u> δ-<u>*neighborhood*</u> *of* a *is the interval* $(a - \delta, a + \delta)$.

The <u>*closed*</u> δ-<u>*neighborhood*</u> *of* a *is the interval* $[a - \delta, a + \delta]$.

Definition. *An* <u>*interval*</u> *is a subset of* $\mathbb{R}$ *of the form*

$$(a, b),\ (a, b],\ [a, b),\ [a, b],\ (a, \infty),\ [a, \infty),\ (-\infty, b),\ (-\infty, b],\ \textit{or}\ \mathbb{R}$$

Definition 5.24. *Let $E \subseteq \mathbb{R}$ and $f : E \to \mathbb{R}$ be a function.*
Let $I \subseteq E$ be an interval. Let $A_f = \{ f(x) \mid x \in I \}$.

We say that f is <u>bounded above</u> on I if A_f is bounded above in $\mathbb{R}$.
We say that f is <u>bounded below</u> on I if A_f is bounded below in $\mathbb{R}$.
We say that f is <u>bounded</u> on I if f is both bounded below and bounded above on I.

Lemma 5.26. *Let* $f\colon [a,b] \to \mathbb{R}$ *such that* f *is continuous on* $[a,b]$.
For every $c \in (a,b)$ *there exists* $\delta > 0$ *such that* f *is bounded on the closed* δ*-neighborhood* $[c-\delta, c+\delta]$.

Proof. Let $f\colon [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Let $c \in (a,b)$.

By a similar argument:

Lemma 5.27. *Let* $f: [a, b] \to \mathbb{R}$ *such that* f *is continuous on* $[a, b]$.
There exists $\delta_a, \delta_b > 0$ *such that* f *is bounded on the closed intervals* $[a, a + \delta_a]$ and $[b - \delta_b, b]$.

We will also need:

Lemma. *Let* $f: [a, b] \to \mathbb{R}$ *such that* f *is continuous on* $[a, b]$.
Let $I, J \subseteq [a, b]$ *are intervals. If* f *is bounded on* I *and* f *is bounded on* J*, then* f *is bounded on* $I \cup J$.

Theorem 5.25. *Let $f\colon [a,b] \to \mathbb{R}$. If f is continuous on $[a,b]$, then f is bounded on $[a,b]$.*

Proof. Let $f\colon [a,b] \to \mathbb{R}$. If f is continuous on $[a,b]$. Let

We want to show that:

1. $\gamma \in B$ i.e. f is bounded on $[a,\gamma]$.
2. $\gamma = b$ i.e. f is bounded on $[a,b]$.

Claim 1. $\gamma \in B$

Claim 2. $\gamma = b$

Theorem 5.28 (Extreme Value Theorem).
If f is continuous on $[a, b]$, then there exists $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \text{ for all } x \in [a, b].$$

Theorem 5.28 (Extreme Value Theorem).
If f is continuous on $[a, b]$, then there exists $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \text{ for all } x \in [a, b].$$

Proof. Let f be continuous on $[a, b]$. We want to show that f attains its maximum value at some $d \in [a, b]$. (The proof that c exists is similar.) Let

Minimizing sequence.

Compactness.

Continuity.

Conclusion.

Theorem 5.31. (Intermediate Value Theorem)
If f is continuous on $[a, b]$ and $f(a) < y < f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.

Theorem 5.29. *There exists $x \in \mathbb{R}$ such that $x^2 - 2 = 0$.*

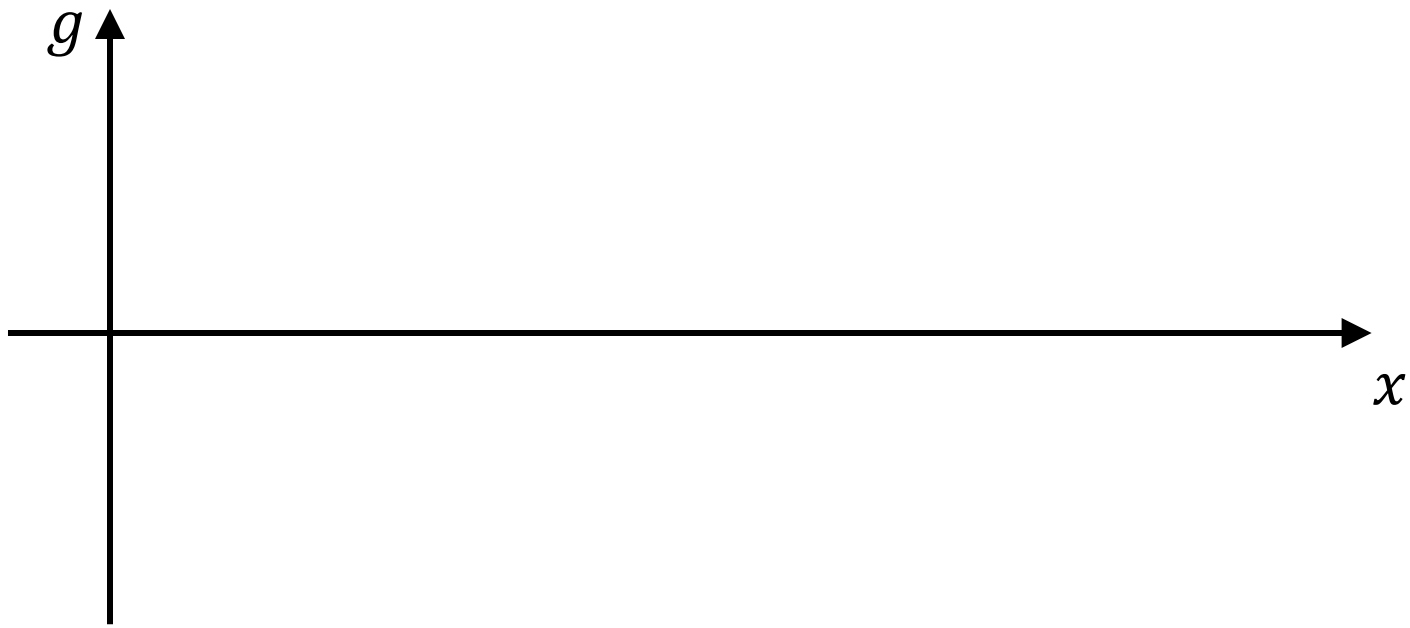
Let $g(x) = x^2 - 2$. Let X be the set of points $x \in [1,2]$ such that $g \leq 0$ on $[1, x]$.

Let $g(x) = x^2 - 2$. Let X be the set of points $x \in [1,2]$ such that $g \leq 0$ on $[1, x]$.

Let $\gamma = \sup X$. We want to show that $g(\gamma) = 0$. Suppose instead that $g(\gamma) > 0$.

By the continuity of $g(x)$ at γ:

For every $\epsilon > 0$ *there exists* $\delta > 0$ *s.t. if* $|x - \gamma| < \delta$ *then* $|g(x) - g(\gamma)| < \epsilon$.



Theorem 5.29. *There exists $x \in \mathbb{R}$ such that $x^2 - 2 = 0$.*

Proof. Let $g(x) = x^2 - 2$ for all $x \in \mathbb{R}$.

We claim that $g(\gamma) = 0$. Notice that since $\gamma \in [1,2]$, g is continuous at γ.

Case 1. Suppose that $g(\gamma) > 0$.

Case 2. Suppose that $g(\gamma) < 0$.

Looking at our proof, let us make the following changes:

- Replace $g(x) = x^2 - 2$ with any continuous function g (relabeled f),
- Replace 1 with any $a \in \mathbb{R}$ such that $g(a) < 0$.
- Replace 2 with any $b \in \mathbb{R}$ such that $g(b) > 0$.

Then by the same proof:

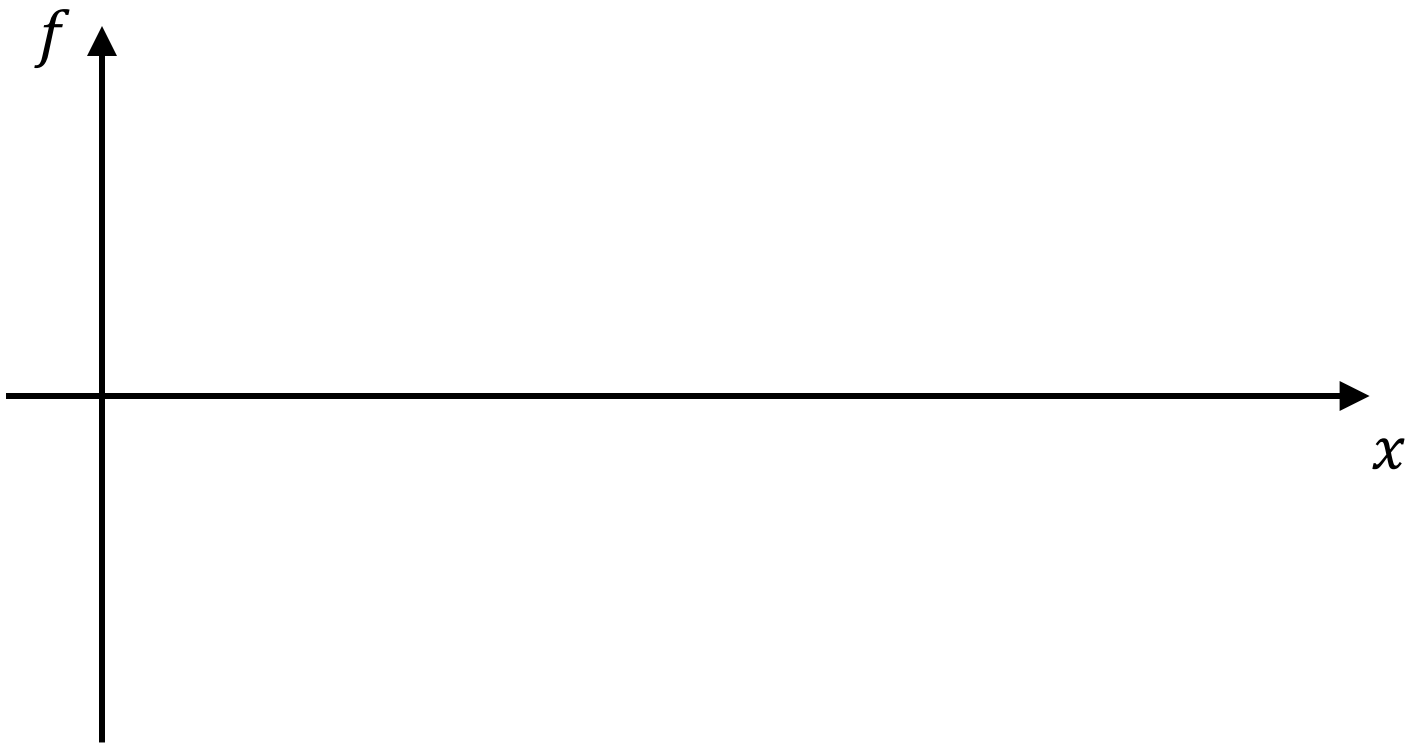
Lemma 5.30. *Let f be continuous on $[a, b]$. If $f(a) < 0$ and $f(b) > 0$, then there exists $c \in [a, b]$ such that $f(c) = 0$.*

Moreover, we showed that:

Since $g(1) = -1$ *and* $g(2) = 2$, *there exists* $c \in [1,2]$ *such that* $g(c) = 0$.

Changing 0 to another value $y \in [-1,2]$ should yield the same result:

For all $y \in [-1,2]$ *there exists* $c \in [1,2]$ *such that* $g(c) = y$.



Theorem 5.31. (Intermediate Value Theorem)
If f is continuous on $[a, b]$ and $f(a) < y < f(b)$, then there exists $c \in [a, b]$ such that $f(c) = y$.

Proof. Let f is continuous on $[a, b]$ and $f(a) < y < f(b)$. Let

Then

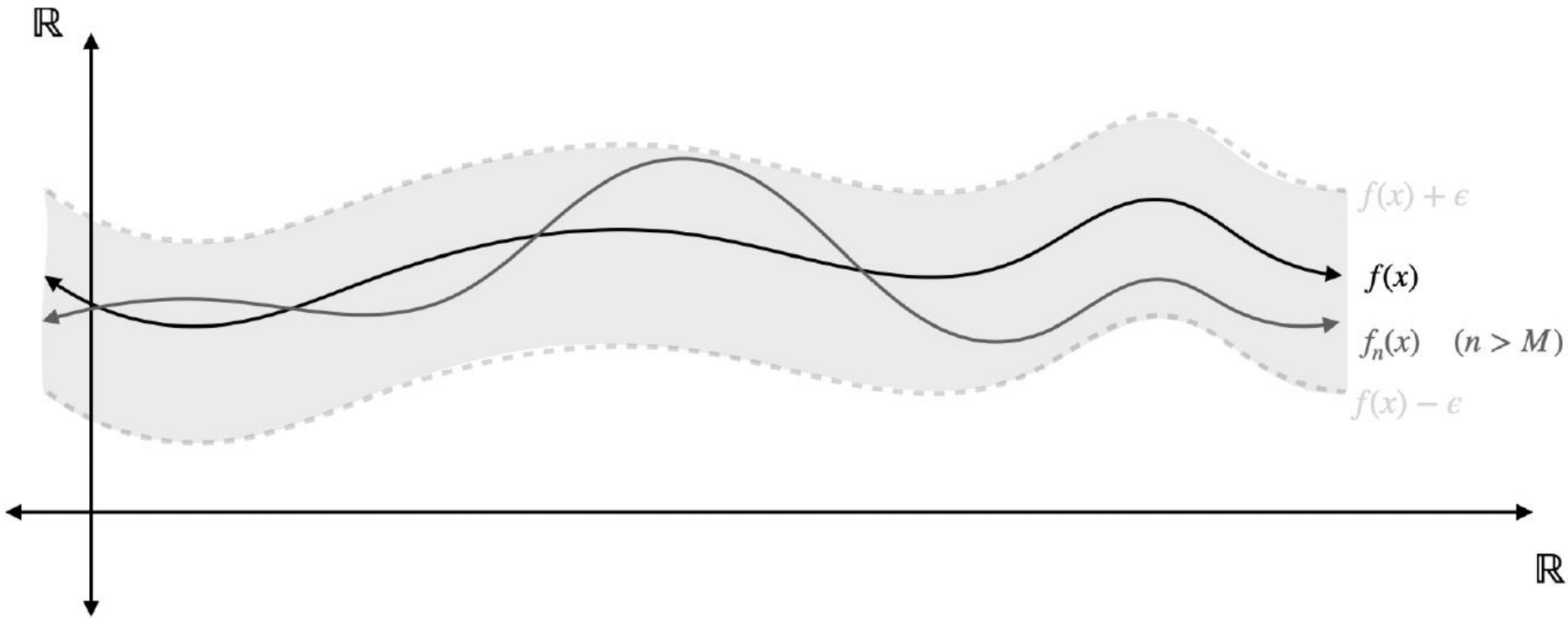
Hence by Lemma 5.30 there exists $c \in [a, b]$ such that $h(c) = 0$. That is

Therefore there exists $c \in [a, b]$ such that $f(c) = y$.

Definition 4.33. *Let $D \subseteq \mathbb{R}$. Let $(f_n(x))$ be a sequence of functions $f_n : D \to \mathbb{R}$ and let $f : D \to \mathbb{R}$ be a function.*

We say that $(f_n(x))$ <u>converges uniformly</u> to $f(x)$ when

$$\forall \epsilon > 0 \; \exists M = M(\epsilon) \in \mathbb{N}^+ \; s.t. \; \forall x \in D \; if \; n > M \; then \; |f_n(x) - f(x)| < \epsilon.$$

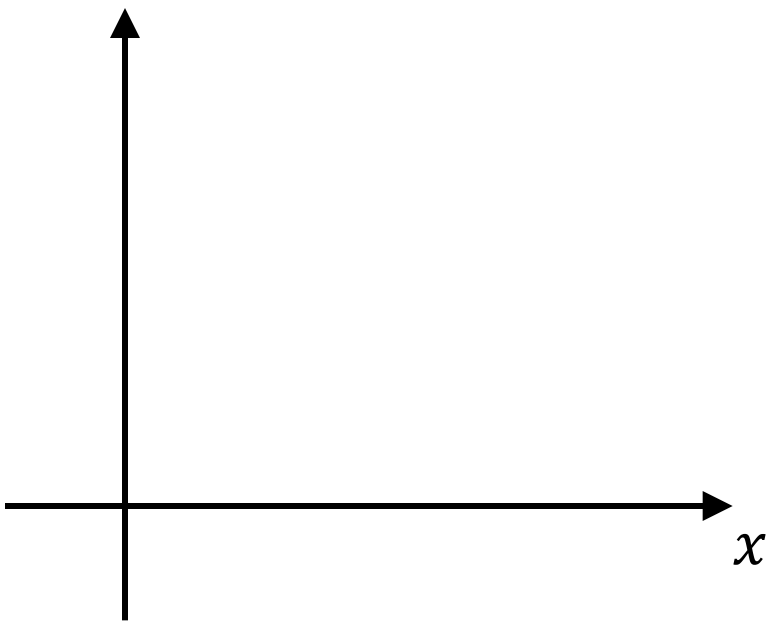


Theorem 5.32. (Uniform Limit Theorem)

Let $(f_n(x))$ *be a sequence of continuous functions on a domain* D.

If $(f_n(x))$ *uniformly converges to* $f(x)$ *on* D*, then* $f(x)$ *is continuous on* D.

Example. $f_n(x) = x^n$ on the domain $[0,1]$.



Theorem 5.32. (Uniform Limit Theorem)

Let $(f_n(x))$ *be a sequence of continuous functions on a domain* D.

If $(f_n(x))$ *uniformly converges to* $f(x)$ *on* D, *then* $f(x)$ *is continuous on* D.

Chapter 6:
Derivatives and Integration

6.1 Derivatives

Example. Let's find the slope of the tangent line to the graph of $f(x) = x^2$ at $x = 3$.

We can approximate the tangent line by the secant lines to the graph passing through $(3,9)$ and (x, x^2).

Definition 6.1. *Let $I \subseteq \mathbb{R}$ be an interval, $f: I \to \mathbb{R}$ be a function, and $c \in I$. We say that f is <u>differentiable</u> at c when there exists $L_c \in \mathbb{R}$ such that*

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = L_c.$$

We call L_c the <u>derivative</u> of f at c.

When f is differentiable at c for all $c \in I$, we say that f is <u>differentiable</u> on I. In this case, we define the function $f': I \to \mathbb{R}$ by $f'(c) = L_c$ for each $c \in I$. That is, $f'(c)$ is the derivative of f at c.

Aside. The derivative of f at c can equivalently be computed as

$$\lim_{h \to 0} \frac{f(c + h) - f(c)}{h} = L_c.$$

Example. $f(x) = x^2$ is differentiable on $\mathbb{R}$.

Example. $g(x) = |x|$ is not differentiable at 0.

Theorem 6.3. *Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$, and $c \in I$. If f is differentiable at c, then f is continuous at c.*

Product Rule. *Let $I \subseteq \mathbb{R}$ be an interval, $c \in I$, and $f, g: I \to \mathbb{R}$ be differentiable at c. Then*

$$(fg)'(c) = f'(c)\, g(c) + f(c)\, g'(c).$$

Example. By induction, we can show that for each $n \in \mathbb{N}$ the derivative of x^n

$$\frac{d}{dx}\, x^n = nx^{n-1}$$

Theorem 6.2. (Linearity of the Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, $c \in I$, and $f, g: I \to \mathbb{R}$ be differentiable at c. Then*

1. $(f + g)'(c) = f'(c) + g'(c)$
2. $(\lambda f)'(c) = \lambda f'(c)$ *for all constants* $\lambda \in \mathbb{R}$.

Aside. Let P_3 be the set of all polynomials of degree at most 3. Theorem 6.2 says that $D: P_3 \to P_3$ defined by $D(f) = f'$ is a linear transformation.

Definition 6.4. *Let* $f\colon [a,b] \to \mathbb{R}$ *be any function and let* $c \in [a,b]$.

We say that f *has a* <u>*local maximum*</u> *at* c *when there exists* $\delta > 0$ *such that*

$$\text{if } |x-c| < \delta \text{ then } f(x) \leq f(c).$$

We say that f *has a* <u>*local minimum*</u> *at* c *when there exists* $\delta > 0$ *such that*

$$\text{if } |x-c| < \delta \text{ then } f(x) \geq f(c).$$

Let $f: [a, b] \to \mathbb{R}$ be differentiable on $[a, b]$. Since f is continuous on $[a, b]$, f attains its maximum (and minimum) values on $[a, b]$.

The relationship between derivatives and extrema comes from:

> *If f is increasing to the left of $x = c$ and decreasing to the right of $x = c$, then f has a local maximum at* c.
>
> *If f is decreasing to the left of $x = c$ and increasing to the right of $x = c$, then f has a local minimum at* c.

Lemma 6.5. *Let* $f: [a, b] \to \mathbb{R}$ *be differentiable on* $[a, b]$ *and let* $c \in (a, b)$. *If* f *has a local maximum (or local minimum) at* c, *then* $f'(c) = 0$.

Intermediate Value Theorem. *Let f be continuous on $[a, b]$ and $f(a) < f(b)$. If $f(a) < y < f(b)$ then there exists $c \in [a, b]$ such that $f(c) = y$.*

The Intermediate Value Theorem answers the question:

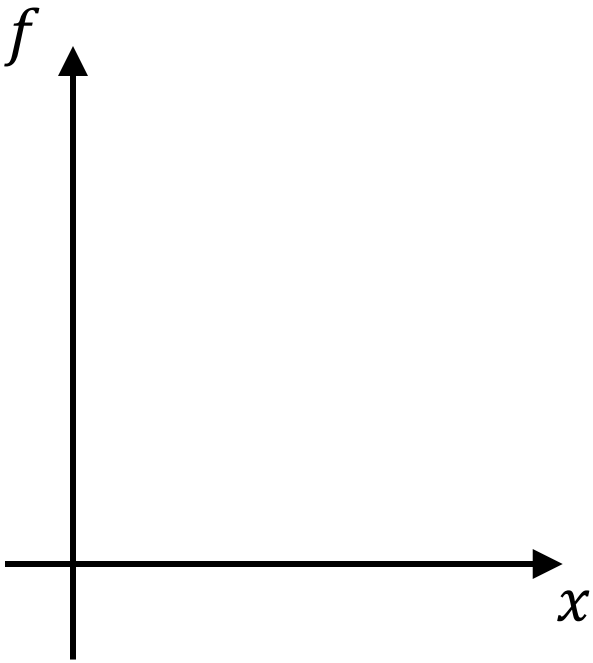
If f is continuous on $[a, b]$, what values is f guaranteed to take on $[a, b]$?

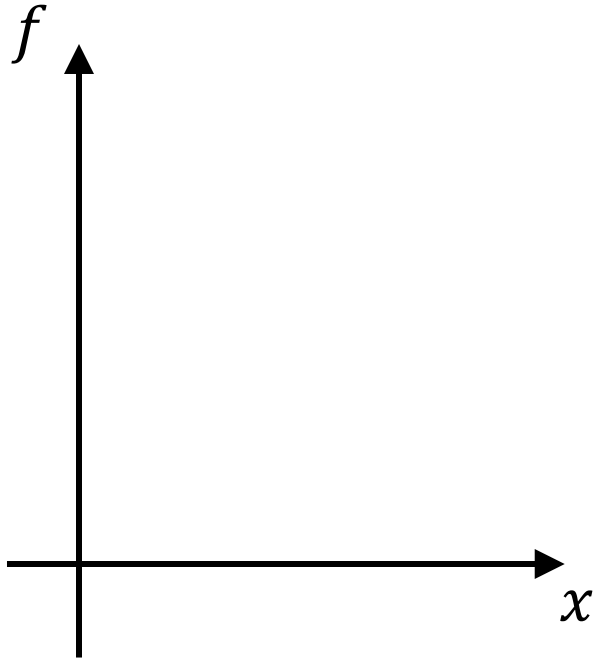
Next we will consider the Mean Value Theorem. It will answer the question:

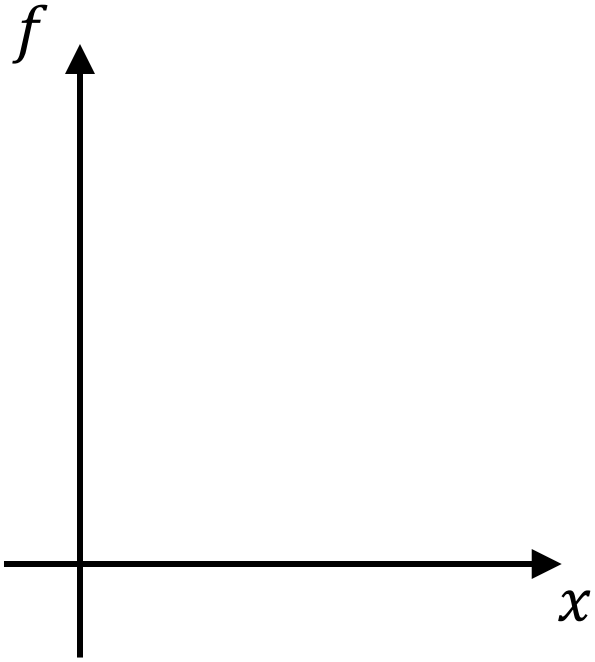
If f is differentiable on $[a, b]$, what values is f' guaranteed to take on $[a, b]$?

Example. For each of the following pictures, try to determine:

If f is differentiable on $[a, b]$, what values is f' guaranteed to take on $[a, b]$?







Theorem 6.6. (Mean Value Theorem) *Let* $f\colon [a,b] \to \mathbb{R}$ *be continuous on* $[a,b]$ *and differentiable on* (a,b). *There exists* $c \in (a,b)$ *such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Example. If a bike travels 30 km in 2 hours, then the bike was going15 km/hr at some point in those two hours.

To see this, let $f: [0,2] \to [0,30]$ s.t $f(t)$ is the distance traveled by the bike (in km) after t hours. Since $f(0) = 0$ and $f(2) = 30$, the Mean Value Theorem tells us

Theorem 6.7. (Rolle's Theorem) *Let* $f: [a, b] \to \mathbb{R}$ *be continuous on* $[a, b]$ *and differentiable on* (a, b).

If $f(a) = f(b) = 0$, *then there exists* $c \in (a, b)$ *such that* $f'(c) = 0$.

Theorem 6.7. (Rolle's Theorem) *Let* $f: [a, b] \to \mathbb{R}$ *be continuous on* $[a, b]$ *and differentiable on* (a, b).

If $f(a) = f(b) = 0$, *then there exists* $c \in (a, b)$ *such that* $f'(c) = 0$.

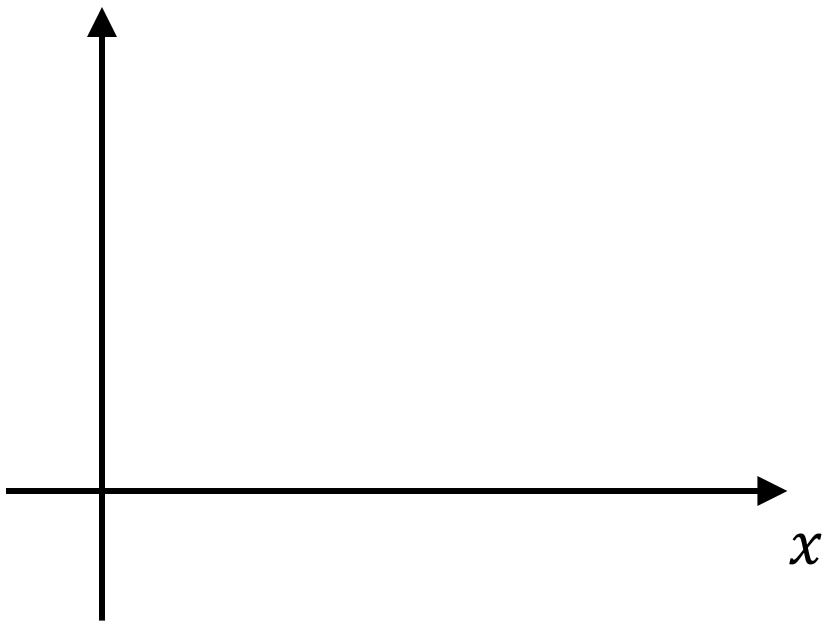
Theorem 6.6. (Mean Value Theorem) *Let* $f: [a, b] \to \mathbb{R}$ *be continuous on* $[a, b]$ *and differentiable on* (a, b). *There exists* $c \in (a, b)$ *such that*

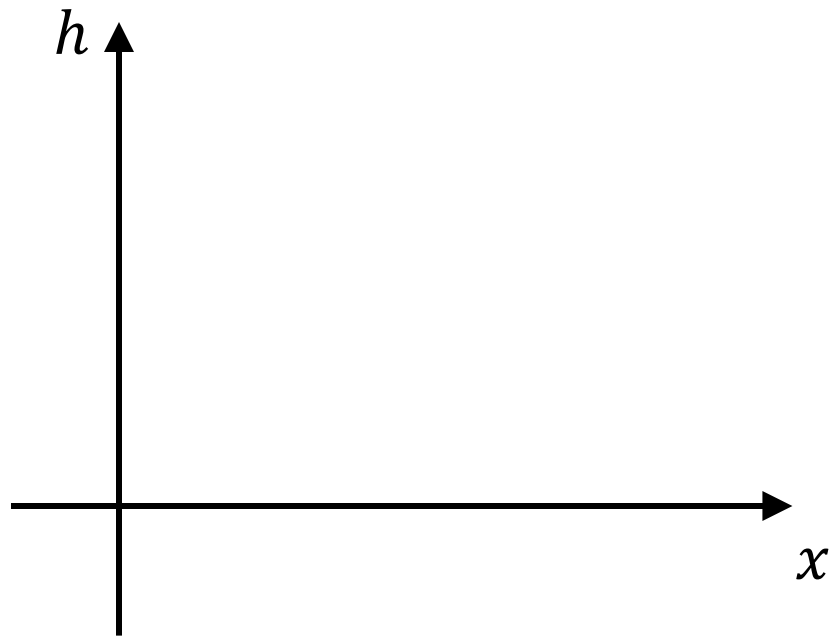
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

To prove the Mean Value Theorem, we need a function $h(x)$ such that

1. $h(x)$ satisfies the hypotheses of Rolle's Theorem including $h(a) = h(b) = 0$;
2. For every c such that $h'(c) = 0$ we have $f'(c) = \frac{f(b)-f(a)}{b-a}$.

The secant line of f between a and b is the graph of the linear function



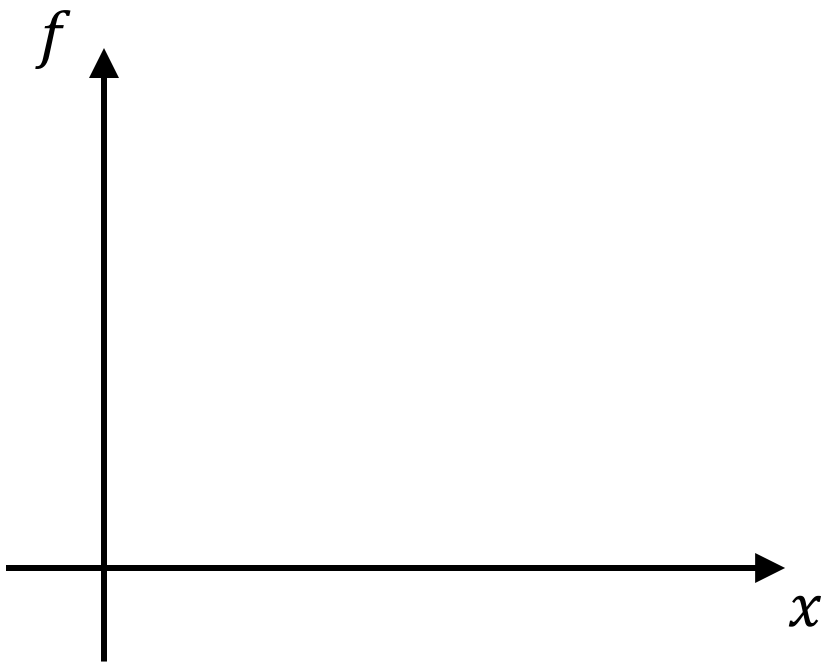


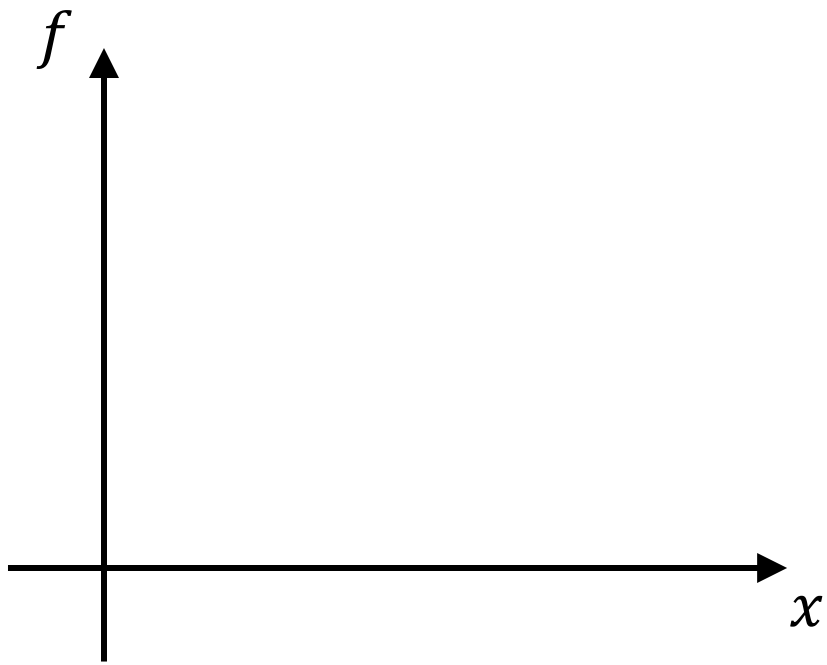
Theorem 6.6. (Mean Value Theorem) *Let* $f\colon [a,b] \to \mathbb{R}$ *be continuous on* $[a,b]$ *and differentiable on* (a,b). *There exists* $c \in (a,b)$ *such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

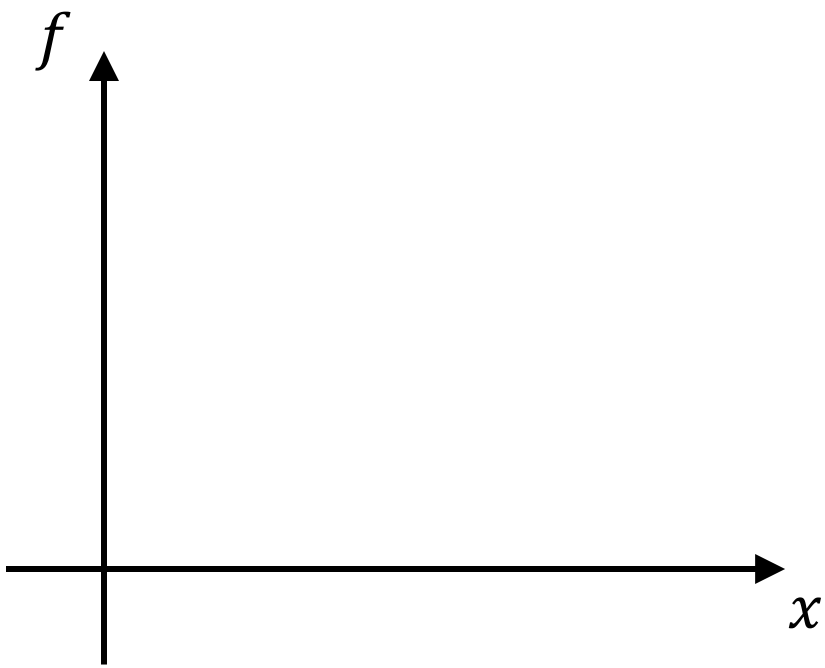
6.2 Integration

Example. Let's find the area under the graph of $f(x) = x^2$ for $3 \leq x \leq 6$.

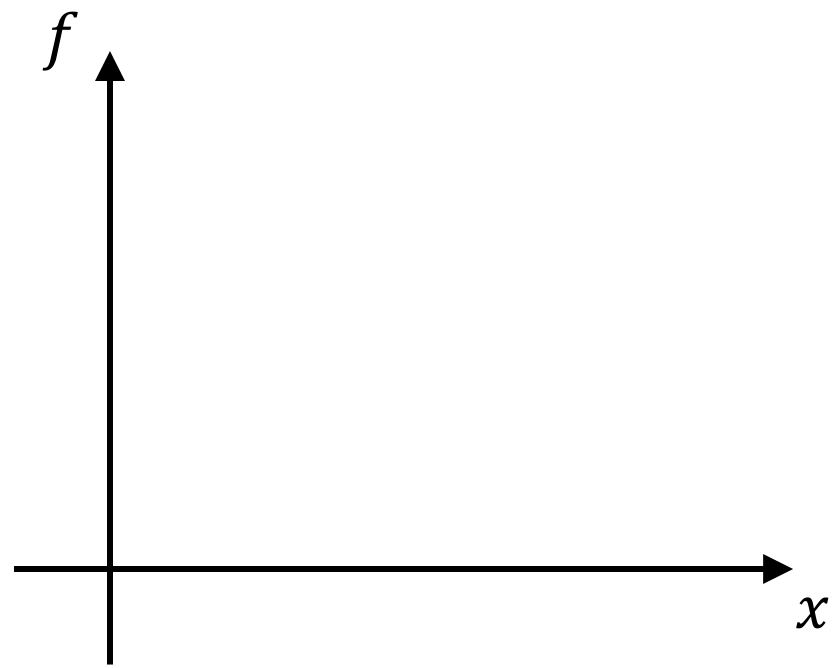




We need not have the intervals have the same length.



For general functions, max/min on each interval need not occur at the end points.



Definition 6.8. *A* <u>*partition*</u> *of* $[a, b]$ *is a set*

$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$$

where

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b.$$

We refer to each interval $[x_{i-1}, x_i]$ *as a* <u>*sub-interval*</u> *of* $[a, b]$.

Example.

Definition 6.9. *Let* $f\colon [a,b] \to \mathbb{R}$ *be bounded on* $[a,b]$ *and let*

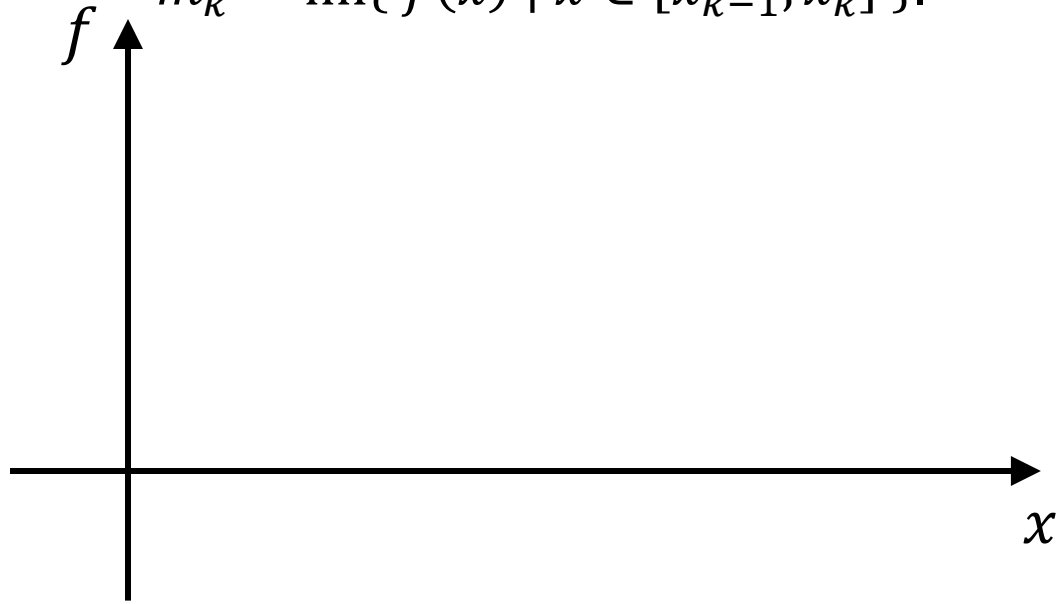
$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$$

be a partition of $[a,b]$. *The* <u>*lower Riemann sum*</u> *of* f *with respect to* P *is given by*

$$L(f,P) = \sum_{k=1}^{n} m_k(x_k - x_{k-1}) = m_1(x_1 - x_0) + m_2(x_2 - x_1) + m_3(x_3 - x_2) + \cdots + m_n(x_n - x_{n-1})$$

where

$$m_k = \inf\{\, f(x) \mid x \in [x_{k-1}, x_k] \,\}.$$



Definition 6.9. *Let* $f\colon [a,b] \to \mathbb{R}$ *be bounded on* $[a,b]$ *and let*

$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$$

be a partition of $[a,b]$. *The* <u>*upper Riemann sum*</u> *of* f *with respect to* P *is given by*

$$U(f,P) = \sum_{k=1}^{n} M_k (x_k - x_{k-1})$$

where

$$M_k = \sup\{\, f(x) \mid x \in [x_{k-1}, x_k] \,\}.$$

Example. Underestimate the area under $f(x) = 25 - x^2$ on $1 \leq x \leq 4$.

Notice that for each sub-interval $[x_{k-1}, x_k]$ we have $m_k \leq M_k$. Therefore

$$L(f,P) = \sum_{k=1}^{n} m_k(x_k - x_{k-1}) \leq \sum_{k=1}^{n} M_k(x_k - x_{k-1}) = U(f,P)$$

This gives us:

Lemma 6.10. *Let $f: [a,b] \to \mathbb{R}$ be bounded on $[a,b]$. For every partition P of $[a,b]$ we have*

$$L(f,P) \leq U(f,P).$$

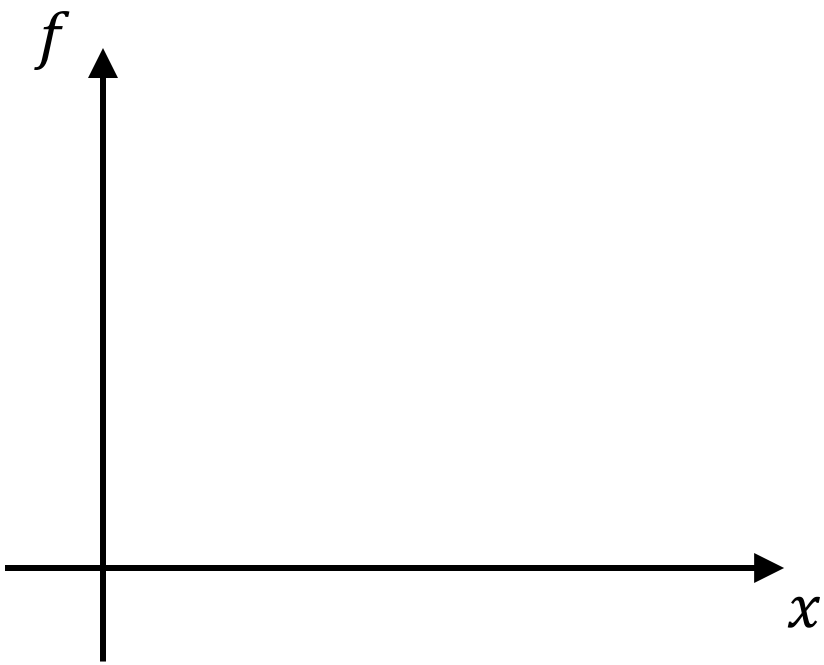
We ideally want: *Let $f: [a,b] \to \mathbb{R}$ be bounded on $[a,b]$. For every partition P, P' of $[a,b]$ we have*

$$L(f,P) \leq U(f,P').$$

Definition 6.11. *Let P and Q be partitions of $[a, b]$.*

We say that Q is a <u>refinement</u> of P when $P \subseteq Q$.

Example. Let $P = \{x_0, x_1, x_2, x_3\}$ be a partition of $[a, b]$ and Q be the refinement obtained by adding a point z between x_1 and x_2.



By using induction, we can include more points in the refinement of P, increasing the value of $L(f, P)$.

We can make the same argument for upper sums. Hence:

Lemma 6.12. *Let $f: [a, b] \to \mathbb{R}$ be bounded on $[a, b]$ and let P be a partition of $[a, b]$. If Q is a refinement of P, then*

$$L(f, P) \leq L(f, Q) \quad \text{and} \quad U(f, Q) \leq U(f, P).$$

Theorem 6.13. *Let $f: [a, b] \to \mathbb{R}$ be bounded on $[a, b]$. If P and P' be partitions of $[a, b]$, then*

$$L(f, P) \leq U(f, P').$$

Definition 6.14. *Let* $f\colon [a,b] \to \mathbb{R}$ *be bounded on* $[a,b]$.

The <u>*lower Riemann integral*</u> *of* f *on* $[a,b]$ *is*

$$L(f) = \sup\{\, L(f,P) \mid P \text{ is a partition of } [a,b] \,\}$$

The <u>*upper Riemann integral*</u> *of* f *on* $[a,b]$ *is*

$$U(f) = \inf\{\, U(f,P) \mid P \text{ is a partition of } [a,b] \,\}$$

Notice that for any partition P of $[a,b]$

$$L(f,P) \leq L(f) \quad \text{and} \quad U(f) \leq U(f,P).$$

As the partitions P get finer, we expect $L(f)$ and $U(f)$ to get closer together so that

$$L(f) = U(f).$$

However, this might not occur, i.e. we might have $L(f) < U(f)$.

Definition 6.15. *Let* $f\colon [a,b] \to \mathbb{R}$ *be bounded on* $[a,b]$.

We say f *is* <u>*Riemann*</u> <u>*integrable*</u> *on* $[a,b]$ *when* $L(f) = U(f)$. *In this case we denote the common value of* $L(f) = U(f)$ *by*

$$\int_a^b f(x)\,dx\,.$$

In other words:

$$L(f) = U(f) = \int_a^b f(x)\,dx\,.$$

Example. Define the *rational indicator function* $\mathfrak{I}: [1,2] \to \mathbb{R}$ by

$$\mathfrak{I}(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Note.

1. We defined the integral of f only using concepts in real analysis, such as sup and inf, and without specifically referring to area.

   One reason is that it is hard to draw good pictures of functions which represent the behavior of the integral in full generality.

   E.g. we often draw graphs of positive continuous functions.

   Another reason is we can readily generalize integration, e.g. of $f: [a, b] \to \mathbb{R}^n$.

2. An integral is the sup/inf of a set. Computing the integral amounts to finding the value of this sup/inf.

3. The rational indicator function $\mathfrak{I}$ is not integrable on any domain $[a, b]$. And so the notation

$$\int_a^b \mathfrak{I}(x)\, dx$$

   is meaningless.

Theorem 6.16. *Let* $f: [a, b] \to \mathbb{R}$. *If* f *is continuous on* $[a, b]$, *then* f *is integrable on* $[a, b]$.

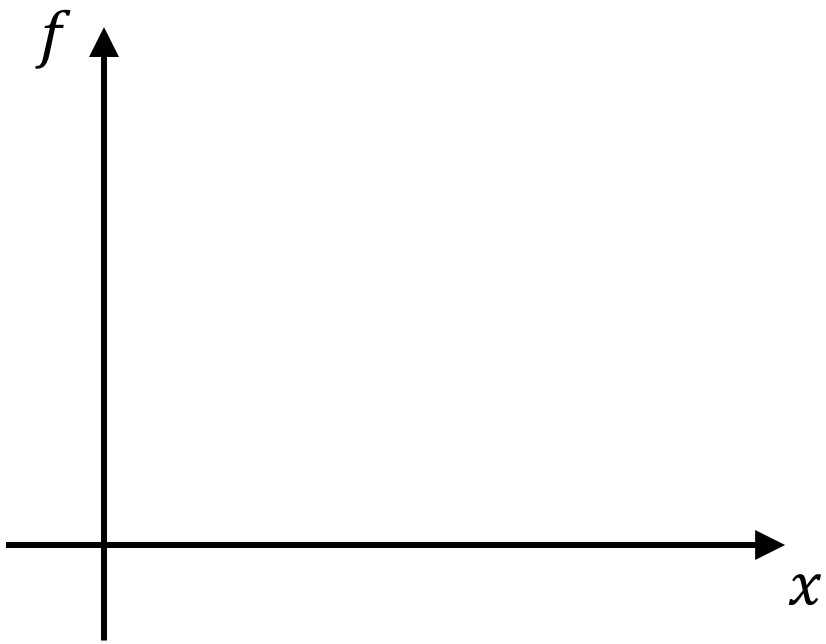
Theorem. *Let* $f, g: [a, b] \to \mathbb{R}$ *be integrable on* $[a, b]$.

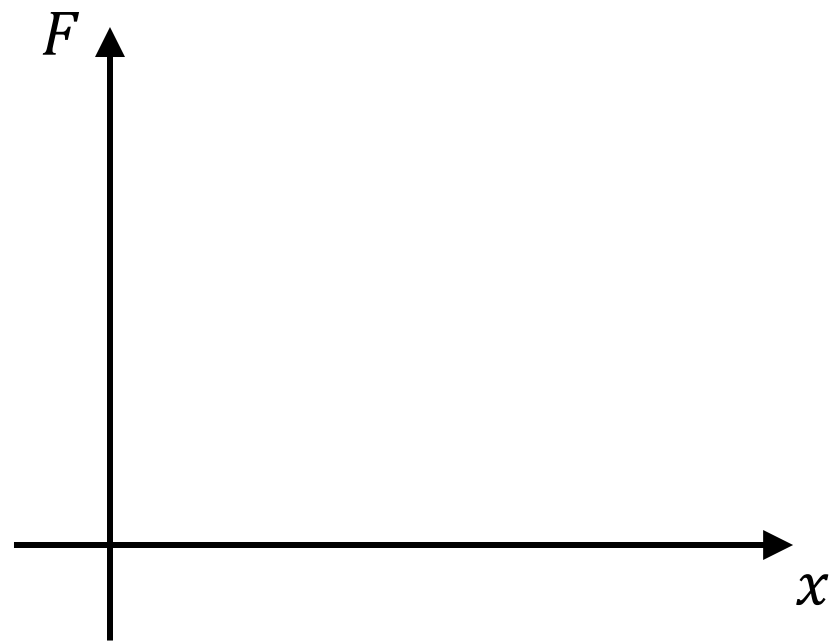
1. $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$ *for all* $c \in (a, b)$.
2. $\int_a^b (f(x) + g(x))\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx$
3. $\int_a^b \lambda\, f(x)\, dx = \lambda \int_a^b f(x)\, dx$ *for all constants* $\lambda \in \mathbb{R}$.
4. $\left| \int_a^b f(x)\, dx \right| \leq \int_a^b |f(x)|\, dx$
5. *If* $f(x) \leq g(x)$ *for all* $x \in [a, b]$, *then* $\int_a^b f(x)\, dx \leq \int_a^b g(x)\, dx$.

Let $f: [a, b] \to \mathbb{R}$ be integrable and $F: [a, b] \to \mathbb{R}$ be differentiable with

$$F'(x) = f(x) \text{ for all } x \in [a, b].$$

Take any partition P and consider the sub-interval $[x_2, x_3]$.





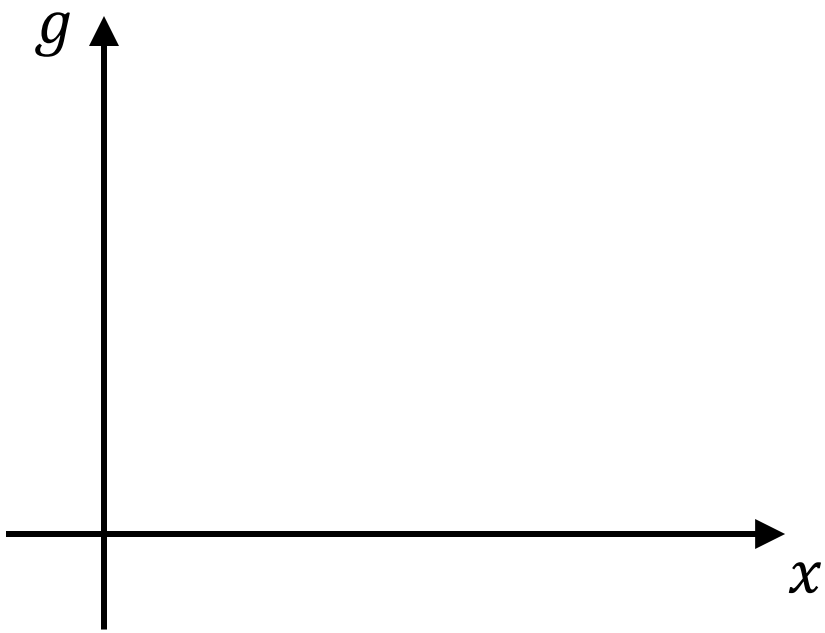
Theorem 6.18. (Fundamental Theorem of Calculus – Part I)
Let $f: [a, b] \to \mathbb{R}$ *be integrable and* $F: [a, b] \to \mathbb{R}$ *be differentiable with*

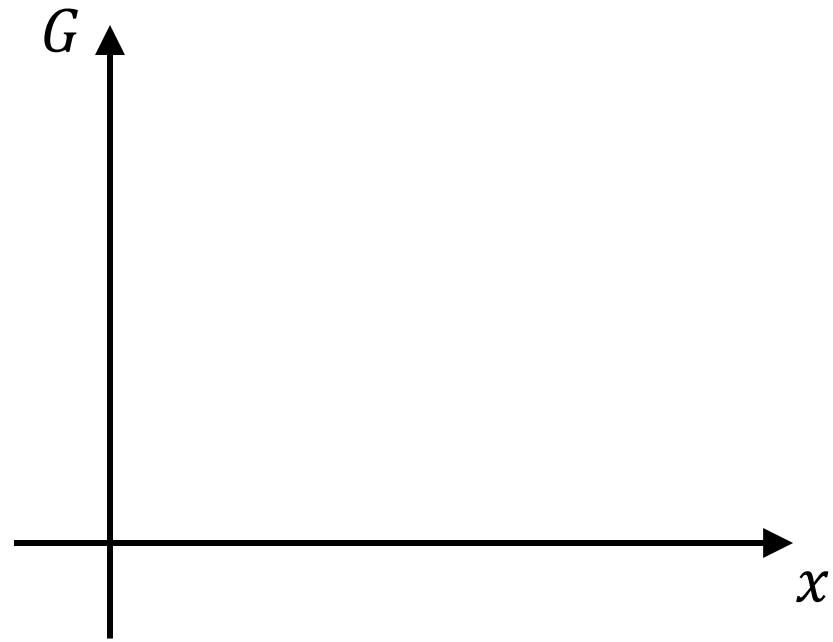
$$F'(x) = f(x) \text{ for all } x \in [a, b]$$

then

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

Example. Let $g\colon [a,b] \to \mathbb{R}$ be the constant function $g(x) = k$.





Theorem 6.18. (Fundamental Theorem of Calculus – Part II)
Let $g: [a, b] \to \mathbb{R}$ *be integrable and* $G: [a, b] \to \mathbb{R}$ *such that*

$$G(x) = \int_a^x g(t)\, dt$$

then

i. *The function* G *is continuous on* $[a, b]$.
ii. *If* g *is continuous at* $c \in [a, b]$, *then* G *is differentiable at* c *and*

$$G'(c) = g(c).$$

Suppose $G: [a, b] \to \mathbb{R}$ is differentiable at c. Consider the line

$$y = G_\ell(x) = G(c) + G'(c)\,(x - c)$$

which passes through $(c, G_\ell(c))$ and has slope $G'(c)$. When x is close to c, we expect $G_\ell(x)$ to be a good approximation of $G(x)$.

Sketch of Proof. Let $g\colon [a,b] \to \mathbb{R}$ be integrable and $G\colon [a,b] \to \mathbb{R}$ such that

$$G(x) = \int_a^x g(t)\, dt\,.$$

Let G is differentiable at c. We want to show that

$$G'(c) = g(c).$$

Chapter 7:
Series

7.1 Series convergence

Definition 7.1. *Let* (a_n) *be a sequence of real numbers. Let* (A_n) *be the sequence with terms*

$$A_n = \sum_{k=0}^{n} a_k = a_0 + a_1 + a_2 + \cdots + a_n.$$

We say that (A_n) *is a* <u>*series*</u>*. The terms* A_n *are called* <u>*partial sums*</u>*.*

We represent the series by

$$\sum_{k=0}^{\infty} a_k$$

Definition 7.2. *Let* (a_n) *be a sequence of real numbers. We say that the series* (A_n) <u>*converges*</u> *to* L *when there exists* $L \in \mathbb{R}$ *such that*

$$\lim_{n\to\infty} A_n = \lim_{n\to\infty} \sum_{k=0}^{n} a_k = L.$$

In this case we write

$$\sum_{k=0}^{\infty} a_k = L.$$

Examples.

$$\sum_{k=0}^{\infty} \frac{1}{4^k} = 1 + \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \cdots$$

Definition 7.3. *Let* (a_n) *be a sequence of real numbers. We say that*

$$\sum_{k=0}^{\infty} a_k$$

<u>*diverges*</u> *when the sequence of partial sums* (A_n) *diverges. That is, for every* $L \in \mathbb{R}$ *the statement* $A_n \to L$ *is false.*

When (A_n) *diverges to* ∞*, i.e. for every* $r \in \mathbb{R}$ *there exists* $M \in \mathbb{N}^+$ *such that if* $n > M$ *then* $A_n > r$*, we write*

$$\sum_{k=0}^{\infty} a_k = \infty.$$

Example.

$$\sum_{k=0}^{\infty} k = 0 + 1 + 2 + 3 + \cdots$$

Definition 7.16. *We say that a series*

$$\sum_{n=0}^{\infty} a_n$$

- *<u>converges absolutely</u> when*

$$\sum_{n=0}^{\infty} |a_n|$$

  converges.

- *<u>converges conditionally</u> otherwise.*

Theorem 7.15. *If a series*

$$\sum_{n=0}^{\infty} a_n$$

converges absolutely, then the series converges.

Theorem 7.3. (Algebra of Series Theorem) *If*

$$\sum_{n=0}^{\infty} a_n = \alpha \quad and \quad \sum_{n=0}^{\infty} b_n = \beta$$

then

$$\sum_{n=0}^{\infty} (a_n + b_n) = \alpha + \beta$$

$$\sum_{n=0}^{\infty} \lambda a_n = \lambda \alpha$$

for every $\lambda \in \mathbb{R}$.

Theorem 7.6. *If the series*

$$\sum_{n=0}^{\infty} a_n$$

converges, then $a_n \to 0$.

Theorem 7.5. (Contrapositive) *If* $a_n \nrightarrow 0$*, then the series*

$$\sum_{n=0}^{\infty} a_n$$

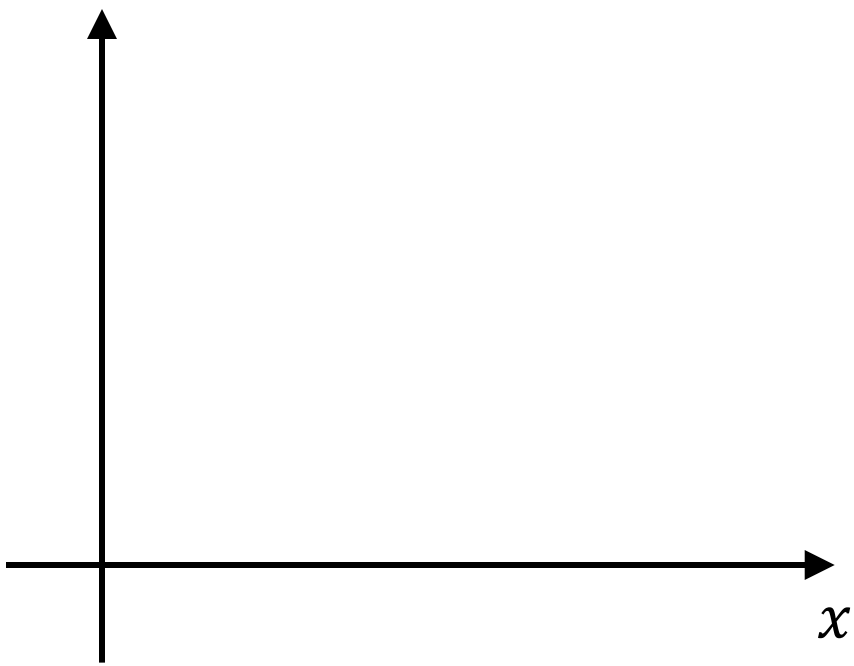
diverges.

Example.

$$\sum_{n=0}^{\infty} \frac{2n^2}{n^2+1}$$

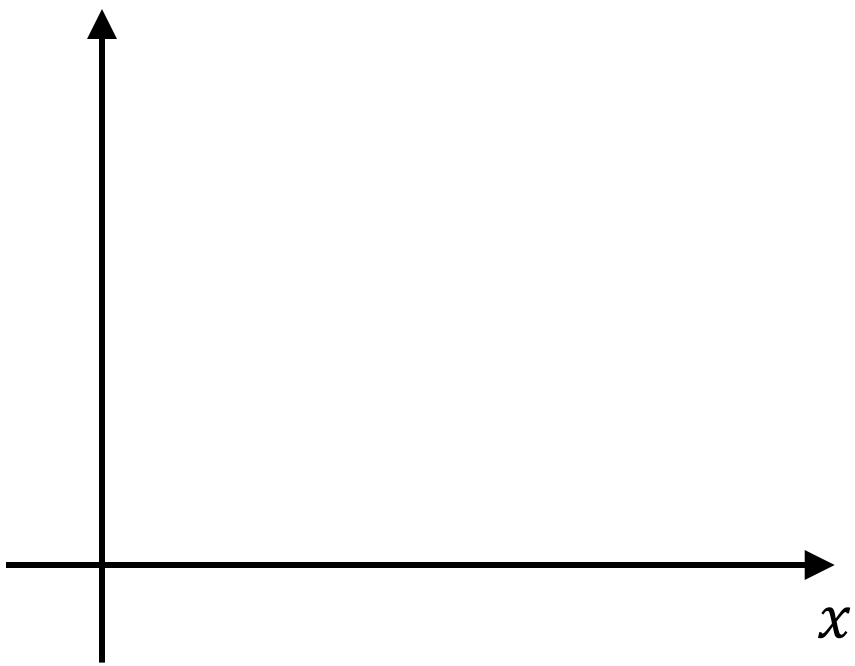
Example.

$$\sum_{n=1}^{\infty} \frac{1}{n}$$



Example.

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Theorem 7.7. (Integral Test)

Let $f: [1, \infty) \to \mathbb{R}$ *be positive, decreasing, and integrable on* $[1, \infty)$.

The series

$$\sum_{n=1}^{\infty} f(n)$$

converges if and only if there exists $L \in \mathbb{R}$ such that

$$\int_1^{\infty} f(u)\, du = \lim_{x \to \infty} \int_1^{x} f(u)\, du = L.$$

Corollary 7.8. (p-series Test)

The series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

- *converges when* $p > 1$*;*
- *diverges when* $0 \leq p \leq 1$.

Definition 7.9. *Let* $a, r \in \mathbb{R}$ *with* $a \neq 0$. *We refer to the series*

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ar^3 + \cdots$$

as a <u>*geometric series*</u>.

Theorem 7.10. (Geometric Series Test)

Let $a \neq 0$ *and* $r \in \mathbb{R}$. *The geometric series*

$$\sum_{n=0}^{\infty} ar^n$$

converges if and only if $|r| < 1$.

Notice that for a geometric series

$$\frac{a_{n+1}}{a_n} = r.$$

The Geometric Series Test tells us that the geometric series converges if and only if this ratio r satisfies $|r| < 1$. This holds true more generally:

Theorem 7.11. (Limit Ratio Test)

Let (a_n) be a sequence such that

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = r.$$

Then if $r < 1$, then the series

$$\sum_{n=0}^{\infty} a_n$$

converges absolutely. If instead $r > 1$, then the series diverges.

Example.

$$\sum_{n=0}^{\infty} \frac{n^2}{2^n}$$

Notice that the Limit Ratio Test tells us nothing about when $|a_{n+1}|/|a_n| \to 1$.

Example.

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges}$$

Consider a sequence (a_k) such that $a_k \geq 0$ for all $k \in \mathbb{N}$. Notice that

$$0 \leq a_0 \leq a_0 + a_1 \leq a_0 + a_1 + a_2 \leq \cdots$$

That is

$$0 \leq A_0 \leq A_1 \leq A_2 \leq \cdots$$

So that the partial sums (A_n) form a monotone increasing sequence.

Theorem 7.12. *Let* (a_k) *be a sequence such that* $a_k \geq 0$ for all $k \in \mathbb{N}$. *The series*

$$\sum_{k=0}^{\infty} a_k$$

converges if and only if there exists $K \in \mathbb{R}$ *such that for all* $n \in \mathbb{N}$

$$A_n = \sum_{k=0}^{n} a_k \leq K.$$

Definition. *Let* (b_n) *be a sequence such that* $b_n \geq 0$ *for all* $n \in \mathbb{N}$. *We call*

$$\sum_{n=0}^{\infty} (-1)^n b_n = b_0 - b_1 + b_2 - b_3 + b_4 - b_5 + \cdots$$

an alternating series.

Theorem 7.13 (Alternating Series Test). *If* (b_n) *is a monotone decreasing sequence such that* $b_n \to 0$, *then*

$$\sum_{n=0}^{\infty} (-1)^n b_n = b_0 - b_1 + b_2 - b_3 + b_4 - b_5 + \cdots$$

converges.

Example.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

Question. When can we treat a series like a (finite) sum?

Example.

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

Example. Let

$$L = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

Suppose that whenever we can rearrange the terms of the series (changing order), the series converges to the same limit L.

Recall:

Definition 7.16. *We say that a convergent series*

$$\sum_{n=0}^{\infty} a_n$$

- *<u>converges absolutely</u> when*

$$\sum_{n=0}^{\infty} |a_n|$$

  converges.

- *<u>converges conditionally</u> otherwise.*

Theorem 7.17. *If*

$$\sum_{n=0}^{\infty} a_n$$

converges <u>*absolutely*</u>*, then every rearrangement of the series converges to the same value.*

Theorem. *If*

$$\sum_{n=0}^{\infty} a_n$$

converges <u>*conditional*</u>*, then for every* $L \in \mathbb{R}$ *there exists a rearrangement of the series which converges to* L*.*

Example. Let us reorder the terms of

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

to get the series to converge to $\pi/4$.

7.2 Power series

Example. Let $a \in \mathbb{R}$ with $a \neq 0$. Consider the following function

$$G(x) = \sum_{n=0}^{\infty} ax^n = a + ax + ax^2 + ax^3 + \cdots$$

Definition 7.18 and 7.28. *Let $b \in \mathbb{R}$ and (a_n) be a sequence. The corresponding* <u>*power series*</u> *at b is given by*

$$\sum_{n=0}^{\infty} a_n(x-b)^n = a_0 + a_1(x-b) + a_2(x-b)^2 + a_3(x-b)^3 + \cdots$$

Definition 7.19. *Let $S \subseteq \mathbb{R}$ be the set of all values x for which the power series*

$$\sum_{n=0}^{\infty} a_n(x-b)^n = a_0 + a_1(x-b) + a_2(x-b)^2 + a_3(x-b)^3 + \cdots$$

converges. We call S the <u>*interval of convergence*</u> *of the power series.*

When S is not bounded, we say the power series has an <u>*infinite interval of convergence*</u>*.*

Theorem 7.20. *The interval of convergence of a power series is one of the following:*

$$\{b\}, \quad (b-\rho, b+\rho), \quad (b-\rho, b+\rho], \quad [b-\rho, b+\rho), \quad [b-\rho, b+\rho], \quad \mathbb{R}$$

We refer to ρ as the <u>*radius of convergence*</u> *of the power series.*

Example.

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Theorem 7.21. (Algebra of Power Series Theorem) *Let*

$$G(x) = \sum_{n=0}^{\infty} a_n (x-c)^n \quad \text{and} \quad H(x) = \sum_{n=0}^{\infty} b_n (x-c)^n$$

be power series at a point $c \in \mathbb{R}$. *Let* S_G *be the interval of convergence of* $G(x)$ *and* S_H *be the interval of convergence of* $H(x)$. *then*

$$G(x) + H(x) = \sum_{n=0}^{\infty} (a_n + b_n)\,(x-c)^n \quad \text{for all } x \in S_G \cap S_H$$

$$\lambda G(x) = \sum_{n=0}^{\infty} \lambda a_n (x-c)^n \quad \text{for all } \lambda \in \mathbb{R} \text{ and } x \in S_G.$$

Theorems 7.22 and 7.23. *Let* $\sum_{n=0}^{\infty} a_n(x-b)^n$ *be a power series which converges to a function* $f(x)$ *at each point* $x \in (b-\rho, b+\rho)$*:*

$$f(x) = \sum_{n=0}^{\infty} a_n(x-b)^n .$$

Then $f(x)$ *is differentiable on* $(b-\rho, b+\rho)$ *and for each* $x \in (b-\rho, b+\rho)$

$$f'(x) = \sum_{n=1}^{\infty} n a_n(x-b)^{n-1} .$$

Moreover, for each $x \in (b-\rho, b+\rho)$

$$\int_0^x f(t)\, dt = \sum_{n=0}^{\infty} \frac{a_n(x-b)^{n+1}}{n+1} .$$

Example. Compute

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

Let $f(x)$ be a function which is at least n-times differentiable at b. Consider the polynomial

$$T_{n,b}(x) = \sum_{k=0}^{n} \frac{f^{(k)}(b)}{k!}\,(x-b)^k$$

$$= f(b) + f'(b)\,(x-b) + \frac{f''(b)}{2!}(x-b)^2 + \cdots + \frac{f^{(n)}(b)}{n!}(x-b)^n$$

where $f^{(k)}$ denotes the k-times derivative of f. We call $T_{n,b}(x)$ the order n <u>Taylor polynomial</u> of f at b.

When $b = 0$, we often write $T_n(x) = T_{n,0}(x)$.

By construction, the first n derivatives of $f(x)$ and $T_{n,b}(x)$ agree at $x = b$.

To see this, suppose for simplicity that $b = 0$ and $n = 3$.

Example. Find the Taylor polynomials of $f(x) = e^x$ at 0.

Example. Find the order 3 Taylor polynomials of $f(x) = \sin(x)$ at 0.

Definition 7.24 and 7.29. *Let $D \subseteq \mathbb{R}$ be an open interval and $f: D \to \mathbb{R}$ be infinitely differentiable at $b \in D$, i.e. the k-times derivative $f^{(k)}(b)$ exists at b for all $k \in \mathbb{N}$. We refer to*

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(b)}{k!} (x-b)^k$$

as the <u>*Taylor series*</u> *of f at b.*

Example. Find the Taylor series of $f(x) = e^x$ at 0.

For every $n \in \mathbb{N}$, $T_{n,b}(x)$ will approximate $f(x)$ near $x = b$. For any particular value of x, the error in this approximation is given by

$$\left|T_{n,b}(x) - f(x)\right|.$$

Ideally, as n gets large the approximation improves and this error term $\to 0$. In other words, for every $\epsilon > 0$ there exists $M \in \mathbb{N}$ such that if $n > M$ then

$$\left|T_{n,b}(x) - f(x)\right| < \epsilon.$$

Theorem 7.25 and 7.30. (Taylor's Theorem)

Let $n \in \mathbb{N}$ and $f: I \to \mathbb{R}$ such that $f, f', f'', \ldots, f^{(n+1)}$ are defined at each point in I. For every $x \in I$ with $x \neq b$ there exists c between b and x such that

$$f(x) - T_{n,b}(x) = \frac{f^{(n+1)}(c)}{(n+1)!}\,(x-b)^{n+1}.$$

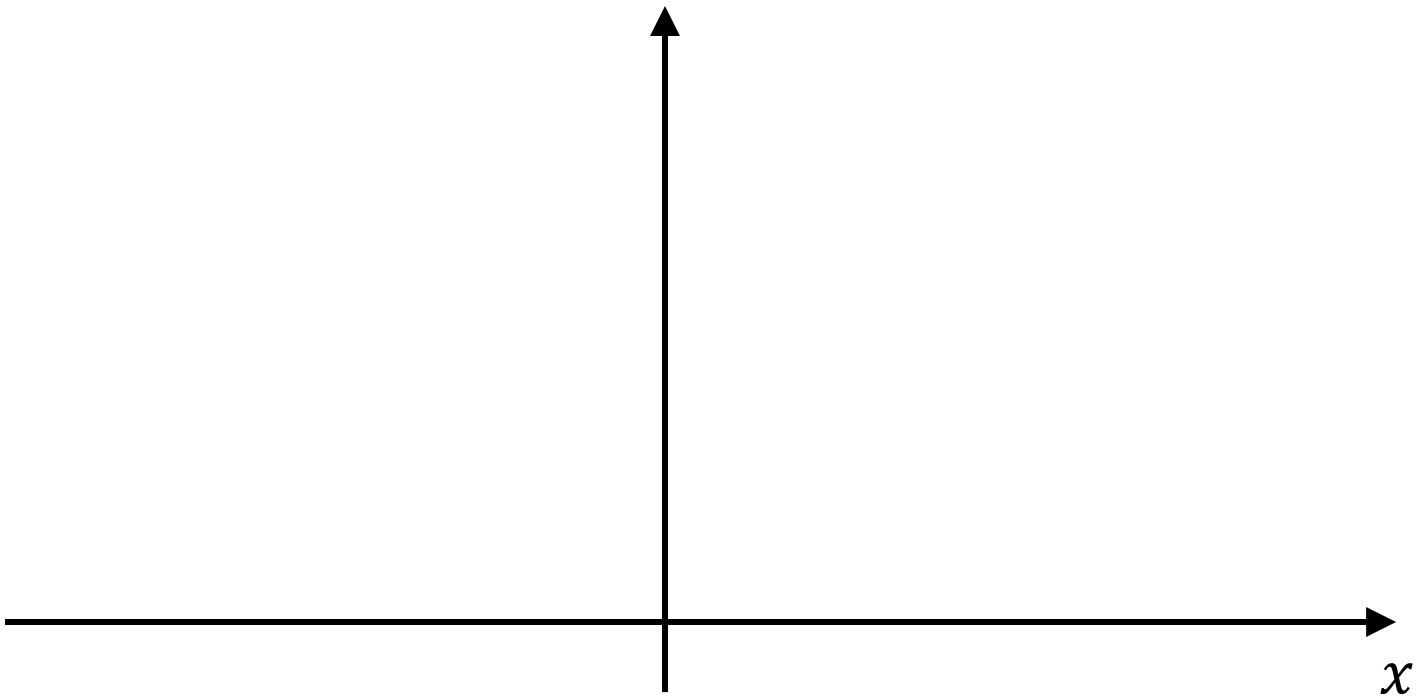
Example 7.26. Prove that

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

for all $x \in \mathbb{R}$.

Example. Consider the function $f: \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} e^{-1/x} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$



Example. Approximate $f(x) = \sin(x)$ by a order 3 Taylor polynomials at 0.

Theorem 7.27 and 7.31. (Remainder Estimate Theorem)

Let $n \in \mathbb{N}$ *and let* $f: I \to \mathbb{R}$ *s.t.* $f, f', f'', \ldots, f^{(n+1)}$ *are defined at each point in* I.

Let $x \in I$ *with* $x \neq b$. *If there exists a constant* $M > 0$ *such that* $\left|f^{(n+1)}(t)\right| \leq M$ *for all* t *between* b *and* x, *then*

$$\left|f(x) - T_{n,b}(x)\right| \leq \frac{M}{(n+1)!}|x-b|^{n+1}.$$

7.3 Fourier series

Let $f: \mathbb{R} \to \mathbb{R}$ be integrable on $[-\pi, \pi]$ and be 2π-periodic, i.e.

$$f(x) = f(x + 2\pi) \quad \text{for all } x \in \mathbb{R}.$$

For each $n \in \mathbb{N}$ we can approximate f by the partial sums

$$F_n(x) = a_0 + \sum_{k=1}^{n} (a_k \cos(kx) + b_k \sin(kx))$$

where $a_0, a_k, b_k \in \mathbb{R}$ are given by for $k \in \mathbb{N}^+$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx\,, \qquad a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\ \cos(kx)\, dx$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\ \sin(kx)\, dx$$

Note that since f is integrable on $[-\pi, \pi]$, $f(x)\ \cos(kx)$ and $f(x)\ \sin(kx)$ are integrable on $[-\pi, \pi]$ and thus a_0, a_k, b_k are defined.

One can check that

$$1 \quad \text{and} \quad \cos(kx), \sin(kx) \text{ for } k \in \mathbb{N}^+$$

form an orthonormal basis with respect to the L^2-inner product

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, g(x)\, dx$$

That is, for all $k, \ell \in \mathbb{N}^+$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} 1\, dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos^2(kx)\, dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin^2(kx)\, dx = 1$$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} \cos(kx)\, dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(kx)\, dx = 0$$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} \cos(kx) \cos(\ell x)\ dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(kx) \sin(\ell x)\ dx = 0 \quad k \neq \ell$$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} \cos(kx) \sin(\ell x)\ dx = 0$$

The partial sums

$$F_n(x) = a_0 + \sum_{k=1}^{n} (a_k \cos(kx) + b_k \sin(kx))$$

are the L^2-projection of f onto the span of $1, \cos(kx), \sin(kx)$ for $k = 1,2, \dots, n$.

Definition. *Let* $f: \mathbb{R} \to \mathbb{R}$ *be integrable on* $[-\pi, \pi]$ *and be* 2π*-periodic. Let* $a_0, a_k, b_k \in \mathbb{R}$ *be as computed on the prior slides. We refer to*

$$a_0 + \sum_{k=1}^{\infty} (a_k \cos(kx) + b_k \sin(kx))$$

as the <u>*Fourier series*</u> *of* f.

Example. Let f be the 2π-periodic function given by $f(x) = x$ for all $x \in (-\pi, \pi]$. Find the Fourier series of the f.

Example. Let f be the 2π-periodic function given by $f(x) = x$ for all $x \in (-\pi, \pi]$. Find the Fourier series of the f.

Since f is odd, a_n = 0

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x \sin nx \, dx = \frac{-1}{n\pi}[x \cos nx]_{-\pi}^{\pi} + \frac{1}{n\pi}\int_{-\pi}^{\pi} \cos nx \, dx$$

$$\frac{-1}{n\pi}[x \cos nx]_{-\pi}^{\pi} - \frac{1}{n^2\pi}[\sin nx]_{-\pi}^{\pi}$$

Example. Let f be the 2π-periodic function given by $f(x) = x$ for all $x \in (-\pi, \pi]$. Find the Fourier series of the f.

Since f is an odd function

$$a_0 = \int_{-\pi}^{\pi} x\, dx = 0$$

$$a_k = \int_{-\pi}^{\pi} x \cos(kx)\, dx = 0$$

We compute

$$\begin{aligned} b_k &= \int_{-\pi}^{\pi} x \sin(kx)\, dx = 2\int_{0}^{\pi} x \sin(kx)\, dx \\ &= \left[\frac{2}{k} x \cos(kx)\right]_0^{\pi} - \frac{2}{k}\int_0^{\pi} \cos(kx)\, dx \\ &= \left[\frac{2}{k} x \cos(kx) - \frac{2}{k^2}\sin(kx)\right]_0^{\pi} = \frac{2\pi}{k}(-1)^k \end{aligned}$$

Therefore the Fourier series of f is

$$\sum_{k=1}^{\infty} \frac{2\pi}{k}(-1)^k \sin(kx)$$

Theorem. *Let $f: \mathbb{R} \to \mathbb{R}$ be bounded and be 2π-periodic.*

Assume that f is piece-wise continuous, piece-wise monotone, i.e. there is a partition

$$-\pi = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = \pi$$

such that f is continuous and monotone on (x_{i-1}, x_i).

Assume that at each point of discontinuity x of f

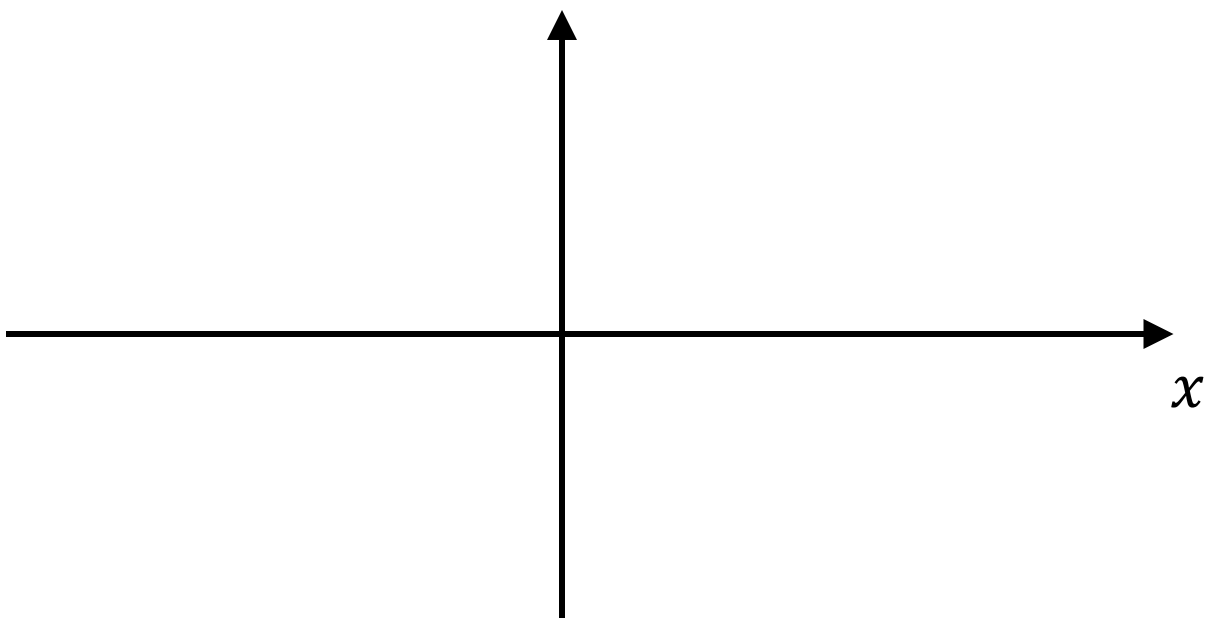
$$f(x) = \frac{1}{2}\Big(\lim_{z \to x^-} f(z) + \lim_{z \to x^+} f(z) \Big).$$

Then the Fourier series of f converges to $f(x)$ pointwise for each $x \in \mathbb{R}$ so that

$$f(x) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(kx) + b_k \sin(kx)).$$

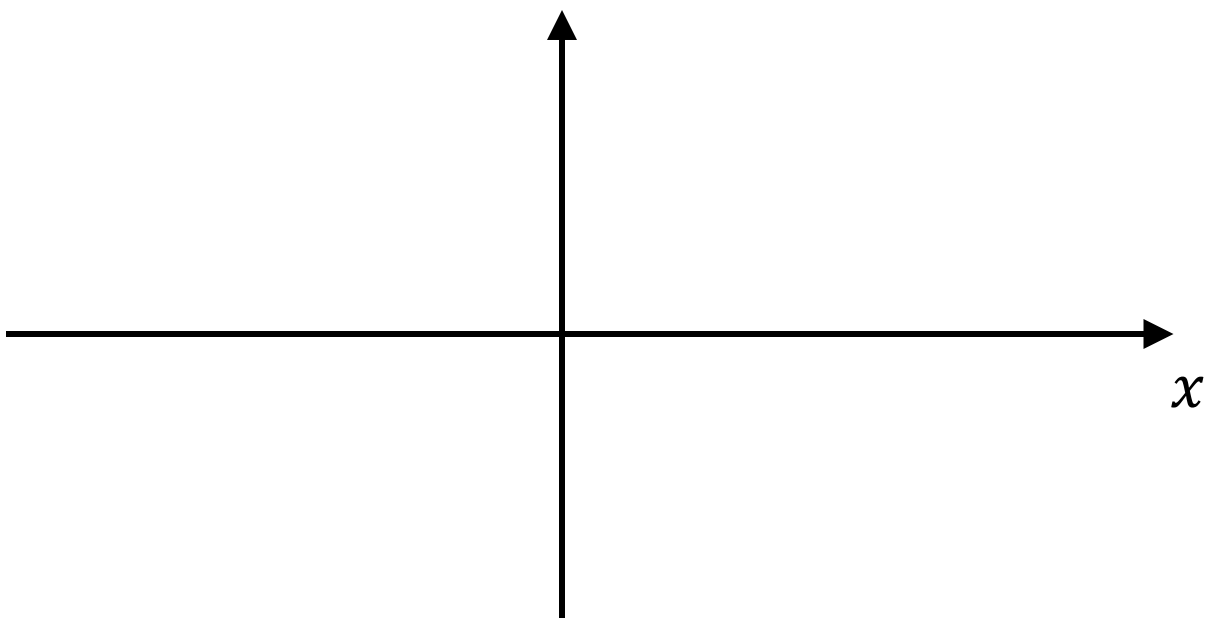
Example. Let f be the 2π-periodic function given by $f(x) = x$ for all $x \in (-\pi, \pi]$. What is the limit of its Fourier series

$$\sum_{k=1}^{\infty} \frac{2(-1)^{k+1}}{k} \cos(kx)\,?$$



Example. Let f be the 2π-periodic function given by $f(x) = x^2$ for all $x \in [-\pi, \pi]$. Use the Fourier series of f to compute

$$\sum_{k=1}^{\infty} \frac{1}{k^2}$$

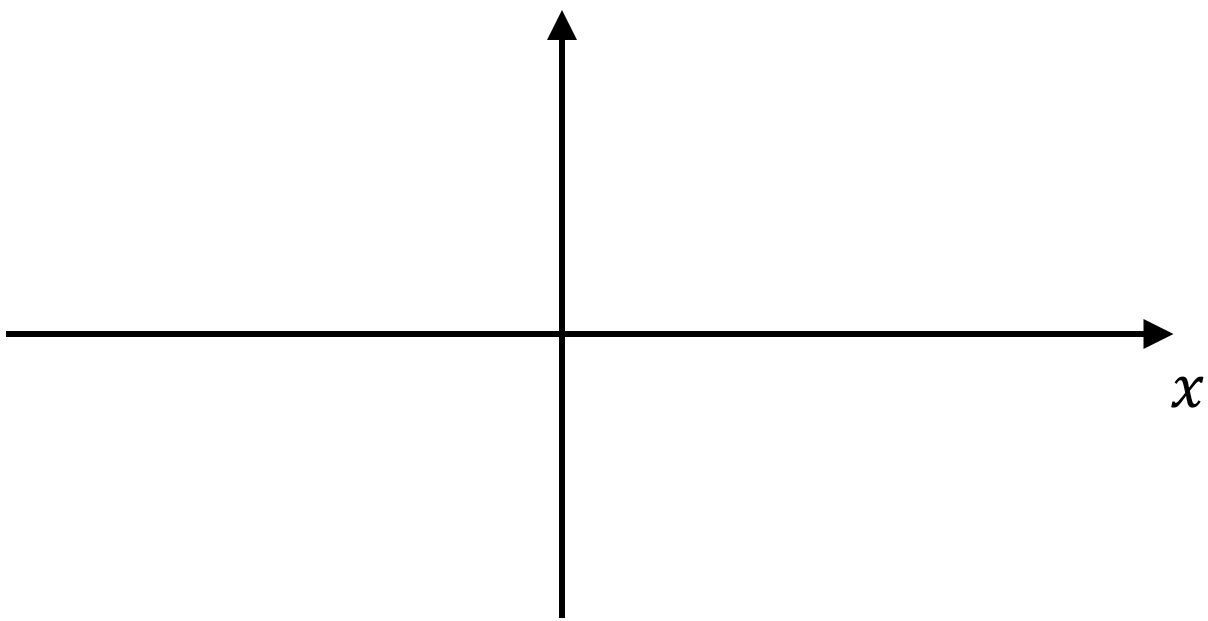


Example. Let f be the 2π-periodic function given by $f(x) = x^2$ for all $x \in [-\pi, \pi]$. Use the Fourier series of f to compute

$$\sum_{k=1}^{\infty} \frac{1}{k^2}$$

Exercise: Show that the Fourier series of f is

$$\frac{\pi^2}{3} + \sum_{k=1}^{\infty} \frac{2(-1)^k}{k^2} \cos(kx)$$



Example. Let f be the 2π-periodic function given by $f(x) = x^2$ for all $x \in [-\pi, \pi]$. Use the Fourier series of f to compute

$$\sum_{k=1}^{\infty} \frac{1}{k^2}$$

The Fourier coefficients of f are

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 \, dx = \frac{\pi^2}{3}$$

$$a_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 \cos(kx) \, dx = -\frac{2}{k\pi} \int_{0}^{\pi} x \sin(kx) \, dx$$

$$= \frac{2(-1)^k}{k^2} - \frac{2}{\pi} \int_0^{\pi} \cos(kx) \, dx = \frac{2(-1)^k}{k^2}$$